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Sensor-driven modeling of complex systems using physics-informed supervised machine intelligence

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ABSTRACT

Modeling the intricate dynamics of complex systems is crucial for addressing global challenges ranging from pandemics to the climate crisis. While conventional first-principles models often struggle, advanced nanosensors employed in arrays and networks provide the high-dimensional and real-time data needed to understand these systems empirically. These datasets are often noisy, incomplete, and manipulated by stochastic, systematic, and nonlinear physical effects, making modeling and prediction challenging. Supervised machine learning (SML) has emerged as a powerful framework to address these challenges by determining explicit input–output relationships, unveiling fundamental system dynamics, enabling accurate prediction, and providing decision support. This review critically examines supervised approaches, including support vector machines algorithms, for sensor-driven modeling of complex systems, with particular emphasis on physics-informed machine learning, where physical laws and domain knowledge are embedded to enhance interpretability, robustness, and generalization. It highlights essential preprocessing methods, including data cleaning, feature engineering, and data fusion, where classical formulations like Kalman filtering, Fourier transforms, and weighted variance minimization integrate sensor physics with computational methods. Further, it discusses core supervised approaches, including regression and classification models, along with their physics-informed variants that integrate conservation laws, stochastic noise models, and network dynamics into learning algorithms as constraints. Through representative case studies across environmental monitoring, biomedical sensing, and industrial diagnostics, their practical utility has been established. Finally, open challenges in scalability, generalization, and integration with physics-informed machine intelligence are outlined with alternate solutions and prospects. This interdisciplinary integration of sensor physics and physics-guided SML represents a powerful paradigm of sensor intelligence for data-driven modeling of complex physical and biological systems.

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I. INTRODUCTION

Complex systems are the backbone of the modern world, comprising dynamical and interconnected networks that span diverse fields, including healthcare, drug discovery, material processing, environmental sustainability, sustainable agriculture, urban planning, and industrial processes. These systems are characterized by heterogeneity, nonlinear interactions, feedback loops, and emergence, whose dynamics cannot be understood by analyzing their individual components in isolation.[1,2] In physics, these systems are often modeled as high-dimensional dynamical systems of coupled, nonlinear differential equations, frequently with stochastic terms to account for inherent noise/unresolved degrees of freedom, and are expressed by the Langevin equation,

$$\frac{dx}{dt} = F(x,t) + \eta(t), \tag{1}$$

where x denotes the state vector representing the multi-dimensional system variables, often obtained through sensor measurements in real-time monitoring frameworks. The term F(x,t) represents the deterministic evolution of the system governed by underlying physical laws such as conservation principles, transport phenomena, or reaction kinetics. The stochastic term $\eta(t)$ captures random fluctuations arising from intrinsic system noise, environmental variability, and measurement uncertainties, which are particularly relevant in sensor-based data acquisition. This formulation highlights the inherent nonlinearity and stochasticity of complex systems, thereby justifying the integration of supervised machine learning (SML) and physics-informed approaches to effectively model, predict, and interpret system behavior from high-dimensional sensor data.[3–10]

Modeling these systems is a practical and scientific imperative to address global challenges related to them, such as climate change, environmental pollution, traffic, pandemics, and antimicrobial resistance. For instance, modeling climatic conditions depends on modeling the interaction of atmospheric, environmental, terrestrial, and oceanic processes to comprehend global warming and its mitigation approaches. It requires coupling the Navier–Stokes equations for fluid dynamics with radiative transfer and thermodynamics. Similarly, understanding the spatial and temporal distribution of environmental factors, such as humidity, temperature, and air pollutants, has been used to model regional modalities to control infectious spread, such as coronavirus disease (COVID-19) [Fig. 1(a)].[11–14] It can be captured by reaction–diffusion models or network-based epidemiological models.

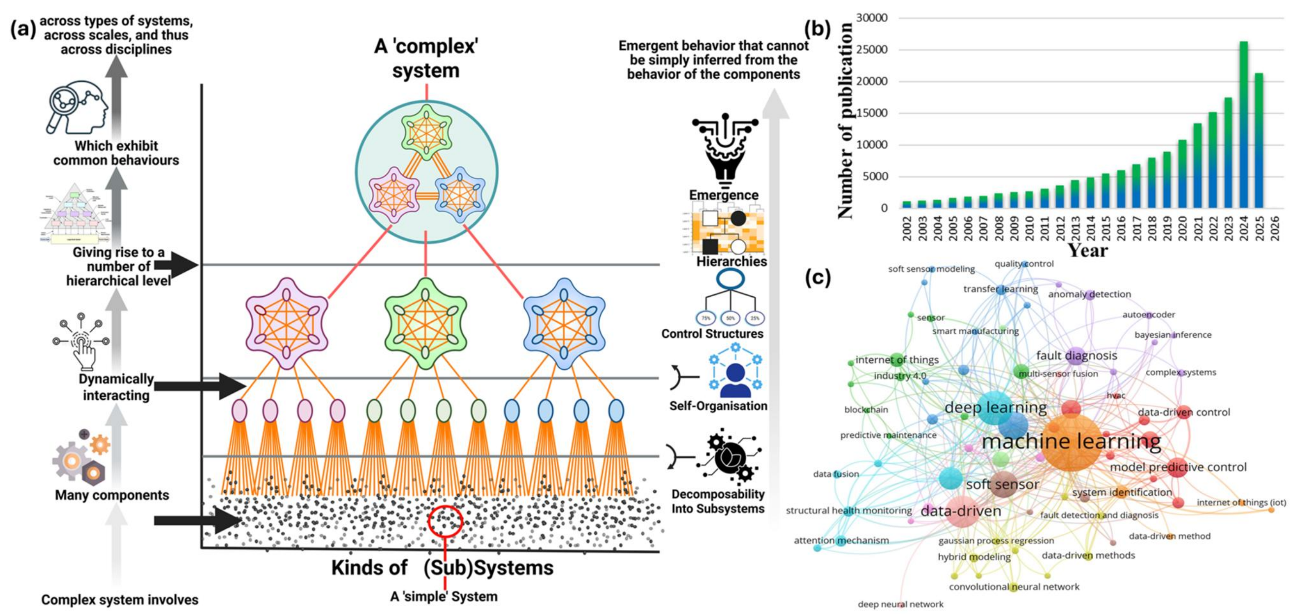

FIG. 1. (a) Schematic representation of "Complex Systems" illustrating their dynamic interactions (relational emergent functions) and hierarchical organization of constituent components or elements, (b) Number of research publications over the years, acquired from ScienceDirect. Search words (sensor-driven, data modeling, and complex systems), and (c) Network visualization on sensors-driven data modeling of complex systems research, the simultaneous appearance of chosen keywords within articles, produced by VOSviewer software. A bibliometric analysis was done to thoroughly assess the published literature from 2016 to 2026. The data were sourced from the Scopus database, focusing on the terms "Sensors," "Data-driven modeling," Complex system, and "AI/ML." A total of 13 944 publications were used for this analysis. VOSviewer software (version 1.6.20) was utilized for the bibliometric analysis. The minimum frequency for a term was set at 10 instances. A map was created using 29 specifically chosen keywords. The study identified eight clusters, represented in red, yellow, green, orange, aqua, purple, pink, and blue, which are particularly significant in this research.

Conventional modeling approaches for complex systems generally rely on abstract frameworks using theoretical, numerical, or equation-based approaches that are often based on simplifying postulations and linear approximations. These approaches, which include agent-based models and mean-field theories, are limited in unveiling the nonlinear, intricate, and emergent behaviors inherent in complex systems. Fundamentally, they frequently fail to examine the full spectrum of spatiotemporal correlations and far-from-equilibrium dynamics.

For example, modeling urban traffic flow by means of static equations is insufficient to unveil real-time changes driven by unexpected disruptions and human behavior. Since the *Lighthill–Whitham–Richards conservation law* works under smooth conditions but breaks down under stochastic human driving behavior and disruptions. Similarly, patient physiology often deviates from linear regression models, instead following nonlinear biochemical kinetics such as the *Michaelis–Menten relation*. These limitations highlight the necessity for more adaptive, robust, and data-driven approaches capable of learning these complex, nonlinear input–output relationships directly from labeled examples, a task for which supervised learning is exceptionally compatible.[15–18]

This requirement has driven the use of data-driven modeling, supported by advanced sensor technologies that generate empirical and multi-dimensional datasets at unprecedented scales. At its core, a sensor is a physical transducer that transforms a particular physical, chemical, or biological event/interaction of interest, the measurand (M), into a detectable signal (S), typically electrical, via a characteristic response function, $S = f(M)$. Modern nanosensors using the quantum confinement effect and size effects offer high sensitivity and resolution. These nanosensors have been integrated into wearable, Internet-of-Things (IoT) enabled, and self-powered architectures, providing real-time data that conventional methods cannot record. This empirical dataset is then fed into artificial intelligence (AI) and machine learning (ML) algorithms, which can recognize patterns, predict outcomes, and help in optimizing processes without requiring explicit postulations about the underlying governing equations of the system. Specifically, supervised learning algorithms are trained on labeled datasets where sensor readings (inputs) are paired with known outcomes (outputs) to learn a predictive mapping function. A particularly powerful fusion of these domains is physics-informed machine learning (PIML), where the learning process is constrained by known physical laws. This paradigm is frequently understood within a supervised learning framework, where the objective (or loss) function, $\mathcal{L}$, in a PIML framework, is usually comprised of a data-fidelity term and a physics-residual term,

$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{phys}. \tag{2}$$

Here, $\mathcal{L}_{data}$ quantifies the divergence with sensor measurements, while $\mathcal{L}_{phys}$ corrects solutions that violate the governing physical equations, with λ being a weighting factor. This integration of nanosensors and SML has created a new paradigm for predictive and informed decision-making in diverse fields.[19–25]

Physics informed machine learning (PIML) is an emerging paradigm that integrates first-principles knowledge with data-driven approaches to model complex systems more effectively. In PIML frameworks, physical constraints are embedded into the learning process through strategies such as incorporating governing equations into loss functions, applying physics-based regularization, and enforcing boundary and initial conditions during training, thereby guiding models toward physically consistent solutions even with sparse or noisy sensor data. By constraining the hypothesis space with known physical laws, PIML enhances interpretability, improves robustness by reducing overfitting to noise and spurious correlations, and enables better generalization to unseen conditions, making it highly suitable for real-world applications such as environmental monitoring, biomedical diagnostics, and industrial process control.[26–30]

The research efforts dedicated to sensor-driven data modeling of complex systems have steadily increased over the years, indicating a sustained rise in academic and industrial interest [Fig. 1(b)]. This trend underscores the expanding role of sensor technologies and data-driven approaches in understanding, optimizing, and managing complex dynamic environments across various domains. A bibliometric analysis of the field [Fig. 1(c)] reveals a convergence of research clusters, primarily linking advances in materials physics (e.g., 2D materials, plasmonic nanostructures) and semiconductor devices with developments in signal processing, sensors, and control theory, with supervised techniques being a major contributor to this expansion.

This paradigm shifts from conventional to sensor-driven data modeling highlights a transformative tool for understanding complex systems. This review focuses specifically on the role of supervised learning in this paradigm. We critically examine the foundational principles, mathematical formalisms, and practical applications of various SML from classical regression to deep neural networks for modeling complex systems with sensor data. We aim to bridge the gap between high-dimensional labeled observational data and actionable physical understanding, highlighting the multidisciplinary nature of these efforts to attain sustainable and adaptive solutions for persistent global challenges.

II. SENSORS-SUPPORTED DATA-DRIVEN MODELING OF COMPLEX SYSTEMS USING SUPERVISED ALGORITHMS

Supervised data-driven modeling uses labeled empirical data gathered from sensors to learn predictive models. For many complex systems, the underlying equations are either unknown or too computationally expensive to solve, making data-driven methods a necessity.[15,31–34] A data-driven model is expressed as a function, $y = f(X;\theta)$, mapping input features X to a prediction y of a known label or outcome, where the parameters θ are learned from sensor data.

The significance of this approach is its efficacy in apprehending real-world dynamics for dynamical and uncertain systems such as turbulent fluid flow or atmospheric phenomena. By integrating sensor arrays with ML algorithms, supervised techniques convert raw sensor measurements into actionable intelligence, allowing unprecedented insight into the behavior of complex systems [Figs. 2(a) and 3(b)].[36–40]

Data preprocessing is the basis for any effective supervised learning model, particularly when dealing with sensor array/network-recorded multidimensional datasets, which can be noisy, incomplete, or heterogeneous. Sensor data inevitably has errors originating from physical sources, including stochastic noise (such as Johnson–Nyquist noise and shot noise in semiconductor devices), systematic effects (sensor drift, calibration errors, and environmental effects), and anomalies (outliers and data gaps due to power failures or transient external interference). Rigorous preprocessing is consequently not

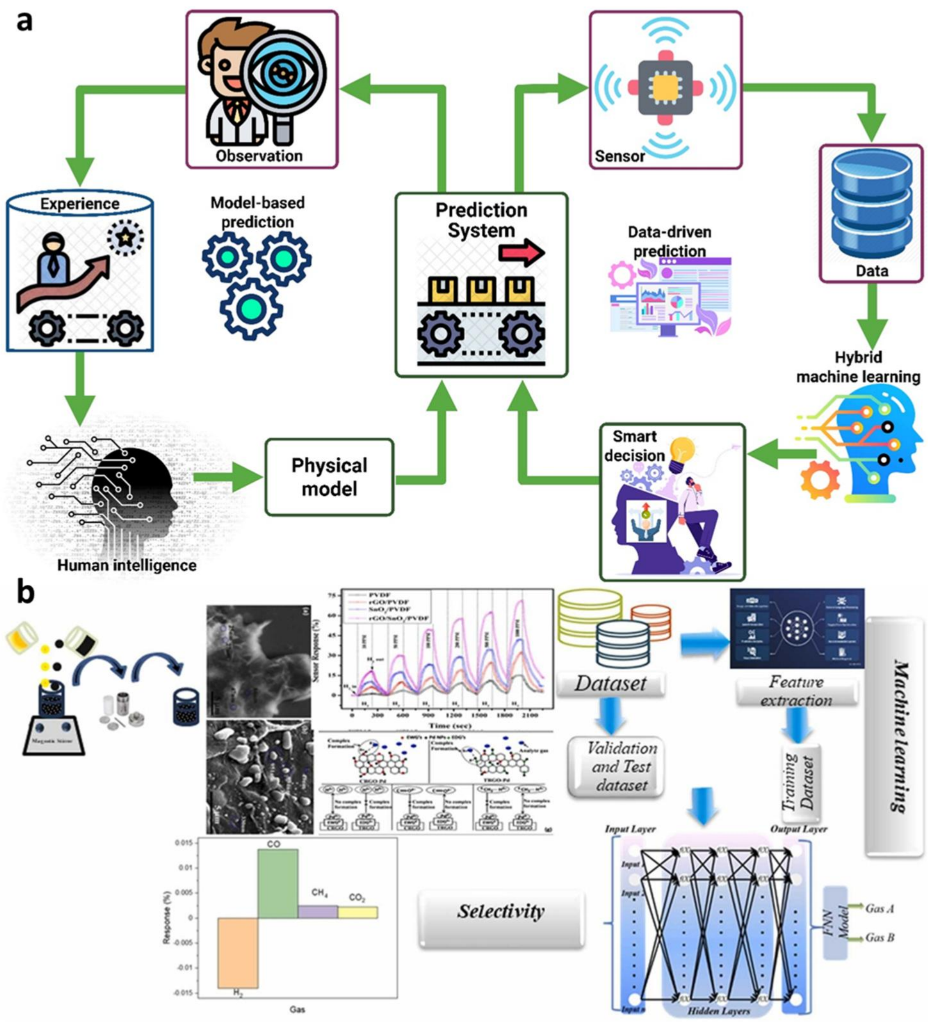

FIG. 2. (a). Sensor-supported data-driven and model-based prediction in complex systems, including different layers consisting of sensor-driven measurement, data, human intelligence, observation, experience, feedback, adaptive, and predictive learning. (b). Schematic showing the workflow from material synthesis and microstructural characterization to gas sensing response and selectivity analysis. Sensor signals are processed through dataset generation, feature extraction, and training of a feedforward neural network (FNN) for gas classification and prediction, demonstrating integration of experimental sensing with machine learning. Reproduced with permission from Pradyumn *et al.*, J. Alloys Compd. **1030**, 180757 (2025). Copyright 2025 Elsevier.[35]

just a computational step but a critical part of the sensor measurement process.[15,41,42] The choice of supervised learning algorithms in sensor-driven modeling depends on data properties such as dimensionality, noise, and temporal dynamics, along with task requirements including prediction or classification. Considerations of model interpretability, scalability, and compatibility with physics-informed constraints further guide selection, ensuring strong and physically consistent modeling of complex systems.[43–48]

A. Data cleaning: The first step toward modeling

Data cleaning provides the accuracy and reliability of sensor-generated datasets. Its primary goal is to mitigate the effects of noise and other corruptions that can obscure the underlying physics and lead to erroneous and biased model predictions.[49–51]

Sensors may fail to record data due to technical problems like power failures, sensor drift, or network issues. Various techniques are used to cater to missing data effectively, including deletion and imputation techniques such as statistical imputation, last observation carried forward, linear interpolation, next observation carried backward, k-nearest neighbor (K-NN) imputation, and multiple imputations. Sensor-generated data are frequently affected by noise due to peripheral disturbances, environmental interferences, or hardware restraints. Simple methods such as the moving average filter act as a numerical low-pass filter, smoothing the data. For a time series x_t, the smoothed value y_t is

$$y_t = \frac{1}{k}\sum_{i=0}^{k-1} x_{t-i}. \tag{3}$$

However, it may have limitations in examining sharp physical transitions. Various advanced techniques are employed to filter noise from data, including Gaussian and median filtering, wavelet transform, and Kalman filtering.[52–56] Kalman filtering is a recursive state estimation method for processing noisy and uncertain sensor data in dynamic systems by integrating prior model knowledge with real-time measurements to yield optimal state estimates. It comprises a prediction step, based on the system model, and an update step, where estimates are corrected using new observations. In sensor-driven frameworks, it enables efficient data assimilation by fuzing

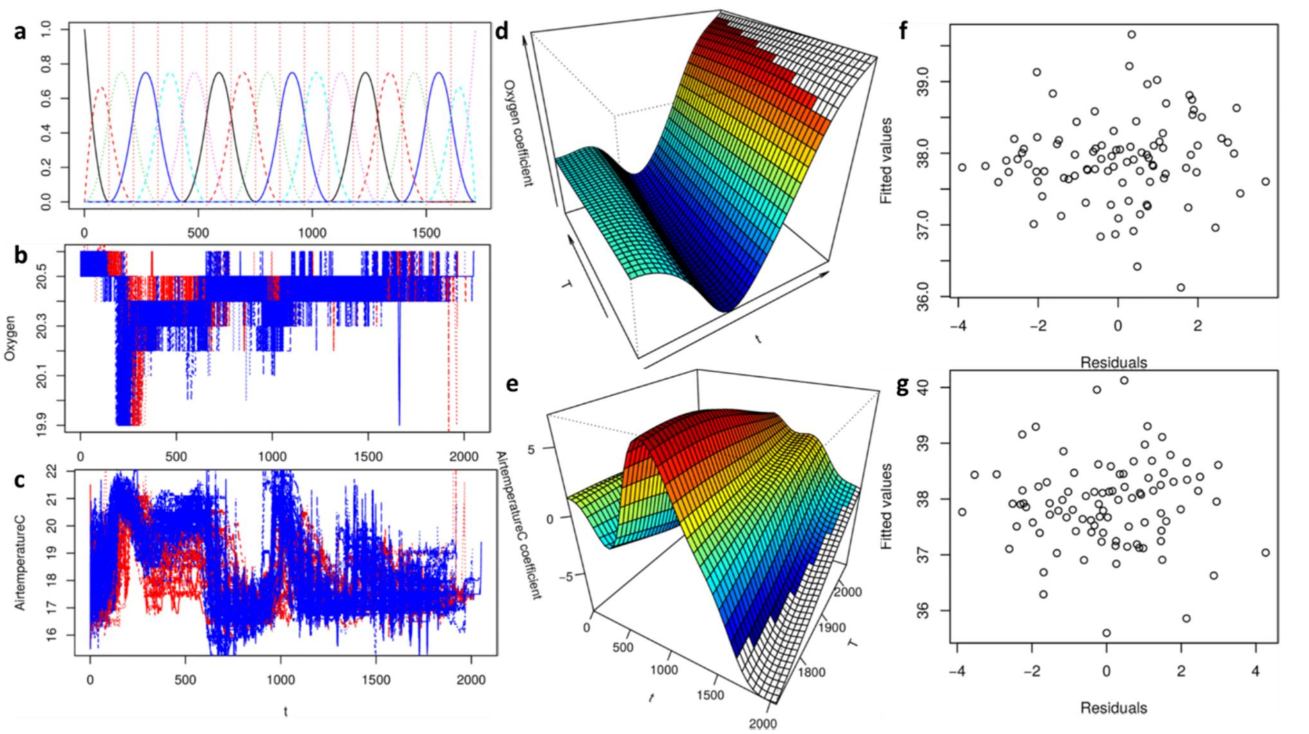

FIG. 3. (a) The primary functions utilized for depicting the temperature curve. The x-axis represents the time index for observations taken every 30 min. (b) Raw oxygen data. The blue lines correspond to growth processes in group G1, and the red lines to group G2, both reflected on the x-axis by the 30 min observation intervals. (c) Raw temperature data. Blue lines indicate growth processes in group G1 and red lines for group G2, with the x-axis showing the time index for each 30 min interval. (d) The calculated coefficient surface from the VDFR utilizing a single functional input (oxygen). Here, the t-axis shows the 30 min time index, while the T-axis indicates the number of days the production process proceeded before harvesting in this experiment. (e) The calculated coefficient surface from the VDFR using a singular functional input (air temperature). The t-axis displays the time index for every 30 min observation interval, and the T-axis represents the duration before harvest for the specific growing trial. (f) Raw residuals for CO_2, (g) Raw residuals for humidity deficit. Reproduced with permission from Panayi *et al.*, PLoS One **18**(1), e0181921 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution (CC BY) License.[103]

multi-sensor inputs while accounting for noise and uncertainties, thereby improving signal quality and reliability. This facilitates robust feature extraction for supervised learning and supports accurate tracking of time-varying states, making it a fundamental preprocessing tool linking physical system modeling with data-driven inference. For instance, the Kalman filter iteratively predicts the next state of the system using the model and then updates its prediction based on the new measurement, providing an optimal estimate for linear systems with Gaussian noise.[52–55] In Kalman filtering, the system state is propagated according to

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad z_t = Hx_t + v_t, \tag{4}$$

where A and H are system matrices, u_t is the control input, and w_t and vt represent the Gaussian process and measurement noise, respectively.

The iterative prediction–correction cycle minimizes the mean square error of state estimation, making it a physics-grounded statistical tool for supervised learning tasks. Another persistent problem with sensor-related data are its duplication. Duplicate data entries can be recorded due to redundant sorting, numerous data sources, or transmission errors. It is necessary to remove this duplicity so that modeling-based outcomes do not misinterpret repetitive information. It can be achieved using techniques such as exact match fuzzing to identify identical data entries and remove duplicates, as well as fuzzy matching, which identifies nearly duplicate data entries using similarity measures.[57,58]

In sensor-driven data modeling, raw data are often generated from various sources with variable units, scales, and distributions. The magnitude differences amongst the features may lead to the biasing of models toward features with larger dominance compared with those with smaller ones. For example, in environmental monitoring setups, pressure might be recorded in Pascals (ranging in thousands), whereas temperature may be measured in degrees Celsius (ranging from 0 to 100). Without proper scaling, data-driven models based on AI/ML may converge slowly or fail to optimize efficiently. Without proper scaling, supervised learning models that rely on distance metrics [like k-NN and support vector machines (SVMs)] or gradient-based optimization [like artificial neural networks(ANNs)] can be biased toward features with larger magnitudes, leading to poor convergence and performance. Consequently, standardization and normalization of sensor data are critical preprocessing steps to improve model stability, convergence speed, and validity.[59–61]

Standardization/Z-score normalization transforms data with a mean of zero and a standard deviation of one,

$$x' = \frac{x - \mu}{\sigma}. \tag{5}$$

In this equation, μ is the mean and σ is the standard deviation of the feature values. It is predominantly beneficial when data follow a normal distribution and is needed for algorithms that adopt a Gaussian distribution, like linear regression analysis. Instead, normalization rescales data to a static range, normally [0,1] or [−1,1], conserving relative differences between values but making all the features contribute evenly to model training,

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{6}$$

It is particularly suitable for models sensitive to absolute magnitudes or that assume a non-Gaussian distribution, like neural networks. In practice, the choice between standardization and normalization is also determined by the physics of the measured quantity (e.g., bounded quantities such as reflectance $0 \leq R \leq 1$ are naturally normalized), while Gaussian noise-dominated signals benefit from Z-scoring. Additionally, log transformations are beneficial for handling skewed distributions and making variations more interpretable. The choice of scaling technique depends on the statistical distribution of the data and the mathematical requirements of the ML model.[62–65]

The data validation step helps to recognize and correct the detected issues, ensuring data integrity. These procedures comprise format validation (ensuring data follows predefined structures like date formats), range validation (examination of whether numerical values fall within expected limits), and uniqueness validation (ensuring fields like IDs are not duplicate). Consistency validation authenticates logical relationships between fields, while completeness checks recognize missing obligatory values. Statistical validation further identifies irregularities by examining data distributions. Automated validation tools, like Python libraries (pandas, Great Expectations), regular expressions for pattern matching, and cross field validation techniques, aid in preserving high data quality. This step reduces errors, improves data-driven decision-making, and ensures compliance with industry standards. By applying efficient data-cleaning approaches, sensor-driven models can produce more accurate, reliable, and significant insights for complex systems analysis.[66–69]

B. Feature engineering for data modeling

Feature engineering is the procedure of converting raw sensor data into significant inputs by creating input variables (features) that are more predictive of the desired output (label). It is crucial in enhancing the accuracy, efficacy, and interpretability of data-driven models for complex systems. The raw data from sensors comprises noise, duplicate data, and irrelevant data, which are efficiently catered to by feature engineering to capture patterns and relationships within the complex system. The crucial steps in feature engineering include feature selection, feature transformation, feature extraction, time-series feature engineering, and domain-specific features. Feature selection is used to identify the most significant variables, those that have the strongest correlation with the target variable, thereby removing redundancy and improving model performance. The selected feature is transformed using mathematical functions (like logarithm and square root) to normalize distributions and improve feature interpretability. The feature extraction derives a new variable from raw data and reduces dimensionality for models based on PCA. Time series feature engineering is used to extract statistical features such as moving averages, trends, and seasonal patterns from sensor-generated data to enhance temporal predictions. It uses the Fourier transform, which decomposes a time-domain signal s(t) into its constituent frequencies $\hat{s}(\omega)$,

$$\hat{s}(\omega) = \int_{-\infty}^{\infty} s(t) e^{-i\omega t} dt. \tag{7}$$

This is used to extract features related to periodicity and resonance, which can be highly predictive of a state of the system or future behavior.[70–76]

C. Data fusion for preprocessing

Data fusion combines data from multiple sensors to create a richer, more informative feature vector for a supervised learning model, thereby improving its predictive accuracy. In sensor-based modeling, data fusion improves reliability, balances individual sensor limitations, and develops enhanced decision-making proficiencies. It includes sensor-level fusion, feature-level fusion, and decision-level fusion. In the first level, all the raw readings/data from multiple sensors, arrays, networks, or sources (like merging data from multiple temperature sensor data) are combined to enhance accuracy and reduce uncertainty.[77–80] For instance, when combining N independent measurements x_i of the same quantity, each with a variance σ_i^2, the optimal fused estimate $\hat{x}$ is a weighted average where the weights are inversely proportional to the variance,

$$\hat{x} = \frac{\sum_{i=1}^{N} \left(x_i / \sigma_i^2\right)}{\sum_{i=1}^{N} \left(1 / \sigma_i^2\right)}. \tag{8}$$

This is the maximum probability estimate for Gaussian noise and ensures that more reliable sensors contribute more to the final output. In the mid-level, relevant features are extracted from different types of sensor data (such as integrating vibration and pressure data for machinery monitoring) and integrated into a unified dataset for further analysis. High-level fusion includes integrating outcomes from multiple supervised models or sensors to derive final decisions, which is frequently employed in AI-driven decision-making systems. These fusion strategies are essential for creating a holistic input for supervised models, enabling a multi-modal understanding of a complex system.[81–86]

III. DATA-DRIVEN PHYSICS-CONSTRAINED SUPERVISED MODELING TECHNIQUES FOR DECODING COMPLEX SYSTEMS

Supervised learning is a paradigm for constructing predictive models of complex systems from labeled data. This approach is used when a dataset,

$$\mathcal{D} = \left\{(x_i, y_i)\right\}_{i=1}^{N}, \tag{9}$$

is available, where x_i is a vector of sensor readings (features), and y_i is the corresponding known outcome (label). The central goal is to learn a function f that approximates the relationship between the features and labels, $y \approx f(x)$. It is done by decreasing a loss function, $L(y,f(x))$, which quantifies the error between the true label and the prediction of the model. For sensor-recorded data, supervised learning algorithms are usually employed for tasks like sensor calibration, sensor drift detection, and error prediction. For example, a sensor-based system that monitors industrial machinery set-up can use a supervised learning technique to envisage machine failures. It will work based on previous sensor data that is labeled with failure events. These techniques include regression techniques, decision trees (DT), random forests (RF), artificial neural networks (ANN), k-nearest neighbors (k-NN), support vector machines (SVM), gradient boosting machines (GBMs), and XGBoost.[87–90] Model performance is evaluated using quantitative metrics such as MSE, root mean squared error (RMSE), and R^2 for regression tasks, and accuracy, precision, recall, F1-score, and AUC for classification. Generalization is assessed through cross-validation and independent test datasets, with additional consideration of robustness to noise and consistency with physical constraints in physics-informed frameworks.[91–93]

A. Regression ML approaches

Regression learnings, such as linear and logistic regression, are crucial in sensor-driven data modeling of complex systems by predicting incessant values and labeling sensor-based events, respectively. Linear regression simulates a direct proportionality between sensor inputs and the target variable, making it the best for forecasting environmental factors like temperature, humidity, or air quality. It is a linear relationship between a set of input features x $(x_1,\ldots,x_p)$ and a continuous target variable y,

$$\hat{y} = w_0 + \sum_{j=1}^{p} w_j x_j = w^T x, \tag{10}$$

where $\hat{y}$ is the predicted value, and the weights "w" are obtained by minimizing the mean squared error (MSE) loss function, typically using optimization methods like gradient descent. It diminishes the error between actual and predicted values through optimization methods such as gradient descent.

However, logistic regression is more appropriate for modeling categorical sensor-driven data, as it predicts the probability of an event happening by mapping input values with a logistic sigmoid function. It models the probability of a binary event ($y = 1$) occurring. The input features are mapped through a logistic sigmoid function to ensure the output is bounded between 0 and 1,

$$P(y = 1|x) = \sigma(w^T x) = \frac{1}{1 + e^{-w^T x}}. \tag{11}$$

It is effective in multi-dimensional and binary classification functions like detecting sensor drift and determining safety thresholds. These regression approaches allow complex healthcare, biomedical, agricultural, and industrial systems to be modeled.[94–100]

For instance, recently, B *et al.*[101] adapted the logistic regression technique to model IoT sensor-gathered data of hospital readmissions for predictive analysis to save healthcare resources and improve patient outcomes. Various physiological parameters (e.g., heart rate, blood pressure, and oxygen saturation) were recorded by IoT sensors to train logistic regression algorithms to recognize sensor data patterns and correlations leading to readmissions. The temporal IoT sensor data allowed dynamic predictions, considering a change in the health of patients over time. These data-driven predictions permit healthcare workers to proactively mediate and provide targeted cures, including prescription alterations, lifestyle guidance, and follow-up consultations to lessen readmission.

On the other hand, Kumar *et al.*[102] predicted the air quality of the three locations in the New Delhi, India region, using linear regression (time series analysis) as an ML technique by modeling sensor-driven data of oxides of carbon, ammonia, and acetone. The model is assessed via four performance measures, including mean absolute error, root mean square error, mean square error, and mean absolute percentage error, which revealed the outcomes from the model to be significant. However, the other prominent factors, such as traffic density, regional constraints, and drastic variations in weather, which may affect the accuracy of the model, can be studied to yield more accurate outcomes.

It also has an application in modeling complex systems for sustainable and precision agricultural practices. Panayi *et al.*[103] used sensor-driven data modeling to quantify the effect of environmental elements on the harvest of *Agaricus bisporus* (button mushrooms) by variable-domain functional regression (VDFR). Unlike conventional methods that necessitate fixed-length statistics, VDFR adapts variable-duration datasets usually found in commercial mushroom farming. Sensor-derived time-series data, such as temperature, humidity, oxygen, and water spraying volumes, are combined into the VDFR model to recognize their dynamic impact on production through diverse growth stages. The outcomes show that optimum oxygen levels diverge through the cycle, with superior levels adversely affecting production in the primary steps but improving it in the absolute stage. It validates the potential of advanced regression methods to model complex, sensor-driven agricultural systems, offering actionable outcomes for improving environmental sustainability [Figs. 3(a)–3(g)].

Kim *et al.*[13] demonstrated the sensor data-driven modeling of complex systems by forming an automatic ventilation control algorithm to optimize indoor environmental quality (IEQ). The method used real-time data from IEQ sensors and inhabitant behavior to train a logistic regression model. The procedure comprises data collection, algorithm development, model assessment using ROC curve examination, and performance validation considering indoor environmental criteria. The logistic regression model, improved with ridge regression, attained a classification accuracy with an area under the curve of 0.865. The algorithm enhanced the compliance rate for ventilation control from 77.6% to 99%, exhibiting the automatization and optimization of sensor-based building ventilation systems based on real-time environmental conditions. It demonstrates a dynamic and empirical method for preserving desired environmental conditions in complex indoor settings.

These algorithms are also used to model sustainable practices. Sriram *et al.*[104] exhibited the sensor data-driven modeling of complex systems by offering an eco-friendly manufacturing forecasting system using IoT and logistic regression. The technique uses industrial IoT sensors to track energy consumption, raw material usage, and pollution levels in real-time. By managing and examining this sensor data, the model is used to predict manufacturing outcomes and related pollution levels. The model aids industries in optimizing resource consumption and decreasing emissions, proposing real-time understandings of environmental effects and permitting alterations to manufacturing to decrease ecological footprints. The method supports active pollution management and confirms environmental compliance by approximating impending pollution levels based on predicted manufacturing situations. By identifying the main causes of pollution, this system improves waste management, operational efficiency, and eco-friendly strategies. It supports sustainable industrial manufacturing, aiding global environmental conservation and responsible manufacturing practices.

While both approaches are computationally proficient and interpretable, linear and logistic regression are not efficient in capturing nonlinearity, making them insufficient for modeling complex systems. They simulate independence between predictors and are sensitive to multicollinearity, resulting in unstable predictions. Moreover, these algorithms depend upon predefined functions, confining their use to determining hidden patterns. While dealing with multi-variate dimensional data, these models are prone to overfitting. These constraints necessitate the use of more advanced models capable of handling nonlinearity and complex feature interactions.

B. Decision trees and random forests ML approaches

Decision trees (DT) and random forests (RF) are powerful supervised learning methods used for both classification and regression. A decision tree operates by recursively partitioning the dataset into purer subsets that are based on feature values. At each node, the algorithm selects a feature and a threshold that best splits the data, creating a hierarchical, tree-like structure. The "best" split is determined by minimizing an impurity metric. For a classification task with K classes, a common metric is the Gini impurity,

$$G = \sum_{k=1}^{K} p_k(1 - p_k), \tag{12}$$

where p_k is the fraction of samples belonging to class k at a given node. This process results in a model that is highly interpretable, making it suitable for direct decision-making. However, they are prone to overfitting, particularly when dealing with noisy or multi-dimensional sensor data. This limitation is addressed by combining random forests with multiple decision trees using an ensemble approach, where each tree is trained on a random subset of the data, decreasing variance and refining generalization. By combining predictions through majority voting (for classification) or averaging (for regression), random forests improve accuracy and strength, making them appropriate for applications such as irregularity detection, sensor fusion, and array/network/system diagnostics.[95,97,105–107]

RF models excel at capturing the complex, nonlinear dependencies between environmental variables and system outcomes. Biswas *et al.*[108] exhibited the application of sensor data-driven modeling in complex agricultural systems by examining the interaction between vegetation indices, climatic variables, and crop yield in Mathura, India. The analysis uses raw sensor data, nine climatic variables, and wheat yield data to examine how these factors influence crop yield. The analysis associate's multiple linear regression and RF models in predicting crop yield variability, exhibiting that the RF model performs better than the MLR model in predictive accuracy. The outcomes reveal the substantial function of integrating sensor-based vegetation and climatic data for more specific agricultural planning and climate adaptation policies through advanced ML models.

Salma *et al.*[109] use sensor data-driven modeling in complex agricultural systems by estimating the radar vegetation index (RVI) to improve the accuracy of leaf area index (LAI) assessment for rice crops. By operating on the RF algorithm, the report reveals the superior functioning of the DOP-derived RVI in predicting LAI, with a high correlation coefficient (R = 0.91) and a low root mean square error (RMSE = 0.25 m^2/m^2). This modeling method demonstrates the use of sensor data to get more accurate and consistent evaluations of crop health and productivity in dynamic and complex agricultural environments. The outcomes emphasize the perspective of combining SAR-based sensors with advanced ML methods to address the complexities of crop monitoring, providing valuable understanding for precision and sustainable agriculture.

The robustness of RF makes it ideal for analyzing complex and often noisy biological data. Wekalao *et al.*[110] demonstrated the role of sensor data-driven modeling in complex biosensing systems by using nanomaterial-based metasurfaces and ML algorithms for high-precision protein recognition. The combination of graphene-perovskite nanocomposite metasurfaces with RF Regression improves sensitivity and recognition accuracy, addressing conventional biosensor limitations. The dual-resonator method, coupled using computational simulations, allows quantum-level sensing efficacy, with spectral shifts attaining sensing responses of 161–226 GHz/RIU. The application of the RF model exhibits an R^2 score of 100%, representing enhanced predictive accuracy in biomolecular recognition. This study highlights the potential of ML-enhanced nanosensors for real-time clinical diagnostics.

Heart diseases are the foremost reason for global mortality, and wearable sensors provide a potential solution for their early detection and monitoring. Rani *et al.*[111] presented a wearable biomedical exemplar devised to predict the existence of heart disease using ECG patterns recorded by an embedded ECG sensor. The RF algorithm was used to examine the data and predict the probability of heart disease, attaining an 88% prediction efficiency. The exemplar provides an economical tool for monitoring heart health, particularly in areas with small doctor-to-patient ratios. By incorporating this technology, elderly individuals and heart patients can get timely alerts, aiding as a basic assistive instrument for personalized heart disease prevention and management. The sensor data-driven modeling of complex physiological systems is also done by combining multisensor technology and ML for noninvasive glucose sensing. A study integrated millimeter-wave, near-infrared, and auxiliary sensors to record physiological factors correlating with blood glucose concentrations.[112] An RF model is trained on data from humans undertaking glucose tolerance examinations, attaining a root mean square error of 21.06 mg/dl as well as a mean absolute relative difference of 7.31%. With 96.1% of predictions lying within clinically acceptable limits, this method emphasizes the potential of ML-enhanced multimodal sensing in advancing noninvasive diabetes monitoring.

Nanomaterial-based sensors show ultra-low detection limits to observe complex systems, but their signal saturation at higher concentrations restricts their dynamic range. To address this, Bian *et al.*[113] employed advanced ML modeling, which involved initiating with linear regression, refining through regression splines, then regression trees, and ultimately RF, which attained an R^2 of 0.8260 by means of out-of-bag error. This enhances the dynamic range of the sensor by 12-fold, allowing precise detection of Hg^{2+} ions from $10^{-14\pm 36}$ M to 1 mM. Moreover, ML-based feature prominence examination offers insights into the FET sensing mechanism, enabling the rational design

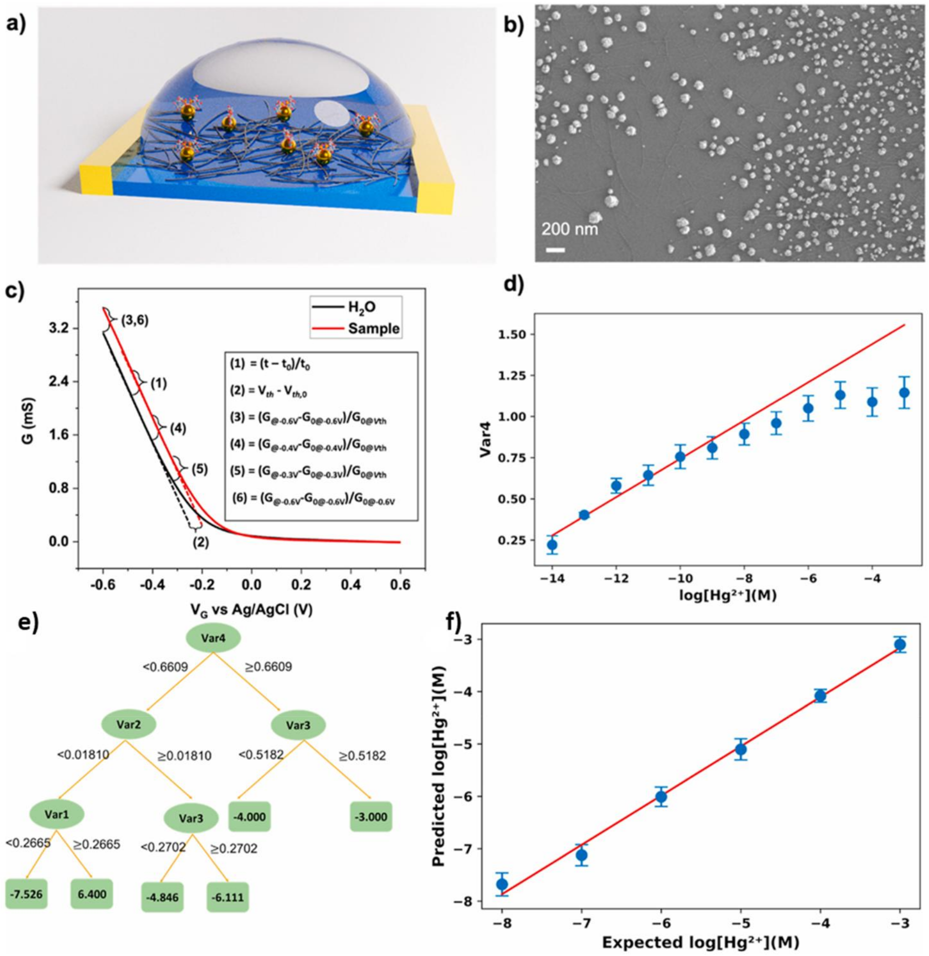

FIG. 4. (a) Schematic of the sensing material and liquid-gated NTFET, where semiconductor-enriched SWCNTs are decorated with Au nanoparticles and functionalized with cysteine and thymine-1-acetic acid for Hg^{2+} detection. (b) SEM image of the functionalized sc-SWCNT network. (c) Definitions and equations of the six extracted features (Var1–Var6). (d) Sensor response (Var4) across the full concentration range, with blue data points (mean $\pm$ error from four devices) and the ideal calibration line (10^{-14} to 10^{-9} M). (e) Structure of the regression tree model. (f) Comparison of random forest-predicted concentrations with expected Hg^{2+} values. Reproduced with permission from Bian *et al.*, Biosens. Bioelectron. **180**, 113085 (2021). Copyright 2021 Elsevier.[113]

of next-generation nanosensors for diverse environmental and biomedical applications [Figs. 4(a)–4(f)].

Proficient and economical wastewater management is critical for sustainable rural development, yet the nonlinearity and hysteresis in sequencing batch reactor methods make conventional modeling methods inadequate. Cheng *et al.*[114] presented a sensor data-driven modeling method applying an RF-based soft sensor to predict chemical oxygen demand (COD) tendencies by means of pH and temperature sensors as cost-effective substitutions to decrease dependence on costly COD sensors. By choosing the uppermost 7 out of 12 input variables, the AI-driven automatic control system improves energy efficiency and carbon emission reduction, substituting a fixed-time regulator. The experimental outcomes exhibit 91.08% COD removal efficacy while saving 24.25% of functioning time and energy. This AI-powered approach improves treatment capacity and offers a low-carbon and economical solution for rural wastewater management, which is complexed with multifold problems.

While RF is a versatile ML algorithm, it has specific limitations when modeling complex systems. It can be prone to overfitting, particularly when working with noisy data, and sometimes be unable to handle multi-dimensional feature spaces efficiently. Moreover, its ensemble nature, concerning multiple decision trees, can result in insufficient interpretability, making it challenging to comprehend the reasoning behind predictions. Additionally, training RF models on big datasets can be computationally costly, resulting in prolonged training times compared to simpler algorithms. In such cases, SVM or ANN can better model nonlinear decision boundaries and often converge faster, making them more effective for modeling complex systems.[115–117]

C. Support vector machine ML approaches

SVMs are prevailing learning algorithms used for categorization and regression functions in sensor data modeling. It is based on determining an optimal hyperplane that divides diverse classes in a multidimensional space, enlarging the fringe between data points. For a binary classification problem, the core idea is to find an optimal hyperplane that separates the data points of two classes with the maximum possible margin. In a p-dimensional feature space, this hyperplane is defined by the equation,

$$w^T x - b = 0. \tag{13}$$

Here, w is a weight vector normal to the hyperplane and b is a bias term. The SVM algorithm finds the optimal w and b by solving a constrained optimization problem: to maximize the margin (proportional to $1/||w||$) subject to the condition that all data points are correctly classified. This translates to minimizing $||w||^2$ under the constraint $y_i(w^T x_i - b) \geq 1$ for each data point (x_i, y_i). They are employed for sensor drift detection, anomaly categorization, and biomedical signal analysis. Kernel functions, including radial basis functions and polynomial kernels, allow SVMs to model nonlinearly separable data. However, SVMs are computationally costly and may necessitate cautious parameter optimization for good performance.[118–120]

The ability of SVMs to effectively handle high-dimensional feature spaces makes them ideal for complex biological data. Surface-enhanced Raman spectroscopy (SERS) integrated with a slippery liquid-infused porous (SLIP) membrane forms a complex system due to its high-dimensional spectral data, nanoparticle aggregation dynamics, and environmental adaptability. Sensor-driven, data-driven modeling using a tree-based support vector machine (Tr-SVM) algorithm allows accurate classification of gene sequences by analyzing subtle spectral variations. A Tr-SVM algorithm was trained by Kang *et al.*[121] on SLIPSERS data to classify 12 gene sequences, such as mecA and intI1 (key antibiotic resistance markers), attaining ∼90% accuracy. This validates the role of AI-driven models to improve label-free biosensing for rapid and high-precision genetic analysis.

The white matter network of the human brain is a complex system described by complex connectivity patterns, dynamic topological arrangement, and nonlinear interactions affecting cognition and behavior. Diffusion tensor imaging-based probabilistic tractography was used by Cheng *et al.*[122] to develop weighted white matter networks for 46 MA-reliant patients and 46 control subjects. Theoretical and graphical examinations showed disrupted network features, comprising reduced small-worldness and modularity, together with augmented nodal efficacy in key brain regions. An SVM classifier trained on these topological features attained ∼98% accuracy, exhibiting the role of AI-driven models in improving the understanding of complex neural disorders through neuroimaging data [Figs. 5(a) and 5(b)].

The application of sensor-driven data-based modeling in the fault analysis of lithium-ion batteries is a complex system necessitating real-time monitoring and predictive maintenance. The intrinsic challenges of noise interference, current fluctuations, and fault identification require robust ML algorithms to improve accuracy and reliability and model this complex system. Yao *et al.*[123] demonstrated an intelligent fault diagnosis technique for LIBs by means of an SVM algorithm. To address noise, distinct cosine filtering with improved truncated frequency was used for denoising. Since conventional covariance matrices are sensitive to current variations, an improved covariance matrix was introduced to enhance robustness. Additionally, grid search optimization is employed to fine-tune the SVM's kernel function and penalty factor for improved fault classification, offering a consistent framework for hierarchical fault management in battery systems and confirming the safety of electric vehicle operations.

The role of SVM modeling in predicting air quality is a complex system perturbed by nonlinear interactions between meteorological and pollutant dispersion factors. An SVM model was reported to estimate PM2.5 concentrations in Bogotá city.[124] The model was trained using a radial basis function kernel. Statistical validation, including root mean square error ($9.302\,\mu g/m^3$) and correlation coefficient (0.654), validated the SVM model's proficiency in short-term air pollution forecasting. This method provides a scalable framework for air quality prediction in other tropical cities, enabling active pollution control approaches.

The single temperature-modulated sensor is a rationalized yet complex system where a single sensor, modulated by dynamic heater waveforms, performs the functionality of an electronic nose for breathomic detection. However, the challenge is interpreting the high-dimensional and nonlinear interactions between breath biomarkers and sensor responses. To address this, ML-driven modeling has been used, using advanced feature selection and a linear SVM classifier to find meaningful patterns from sensor data. Chronic obstructive pulmonary disease (COPD) and air quality index (AQI) are modeled within a unified sensor-driven framework by integrating high-resolution spatiotemporal pollutant data with physiological response variables to resolve exposure–response dynamics across scales. Feature representations are constructed from multi-sensor streams, including concentration fields of PM2.5, PM10, NO_x, CO, and volatile organic compounds (VOCs), their temporal gradients, dispersion patterns, and exposure dose metrics, coupled with clinically relevant respiratory indicators such as spirometric measures (FEV1, FVC), oxygen saturation, and breathing patterns. Derived features further include cumulative exposure indices, lagged exposure effects, frequency-domain descriptors, and higher-order statistical moments to capture non-stationarity and nonlinear interactions. These features are mapped to regression and classification tasks using supervised learning architectures, where PIML imposes constraints derived from advection–diffusion transport equations, chemical transformation kinetics, and established dose–response relationships in pulmonary physiology. Such constraints regularize the learning process, ensuring physically consistent solutions under sparse, noisy, and heterogeneous data conditions. This integrated approach enhances model robustness and generalization across diverse environmental regimes and population cohorts, enabling improved prediction of COPD onset and progression, severity stratification, and quantitative assessment of AQI-driven health risks. Mahdavi *et al.*[125] reported a single temperature-modulated sensor to detect COPD from human breath. The model attained 80.60 % accuracy, 78.79 % sensitivity, and 82.35 % specificity in distinguishing between COPD patients and healthy individuals. These outcomes validate the potential of suitably configured sensors to take comprehensive breath information for COPD detection.

Despite their efficacy in classification tasks in modeling complex systems, SVMs have some limitations, resulting in the adoption of alternative ML models like kNN and ANNs. One main disadvantage of SVMs is their excessive computational complexity, particularly while modeling large datasets, as training entails solving a quadratic optimization problem that scales poorly with cumulative data size. Moreover, SVMs are not good with noisy data and overlapping class distributions and fail to define an optimal hyperplane. The dependence on kernel functions for nonlinear classification also has challenges, as choosing the suitable kernel is frequently heuristic and necessitates extensive tuning. Finally, as a native binary classifier, handling multi-class problems requires strategies like "one-vs-one" or "one-vs-rest," which can be less efficient and elegant than intrinsically multi-class models like ANNs.[126–129]

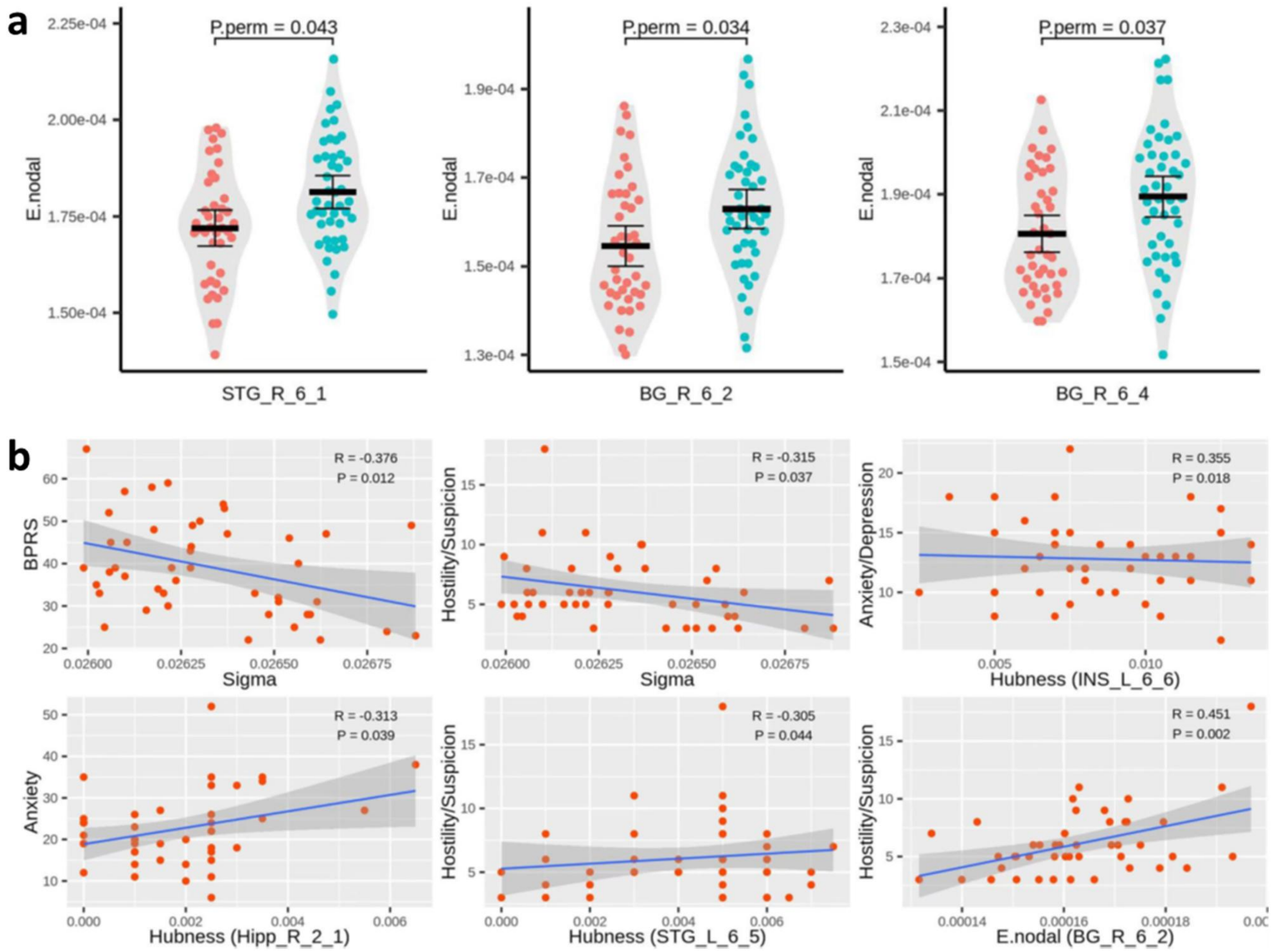

FIG. 5. (a) Substantial differences in nodal efficiency were observed in specific brain regions between methamphetamine-dependent patients and healthy controls. STG_R_6_1 represents the weighted nodal efficiency of the right medial superior temporal gyrus, BG_R_6_2 indicates the weighted nodal efficiency of the globus pallidus, and BG_R_6_4 reflects the weighted nodal efficiency of the ventromedial putamen. (b) The study examines the relationship between weighted topological attributes and HAMA or Brief Psychiatric Rating Scale (BPRS) scores (either total or factor scores) in methamphetamine-dependent individuals. Sigma denotes the small-worldness scalar, INS_L_6_6 refers to the hubness scores of the left dorsal dysgranular insula, Hipp_R_2_1 denotes the hubness scores of the rostral hippocampus, STG_L_6_5 indicates the hubness scores of the left superior temporal gyrus caudal area, BG_R_6_2 reflects the weighted nodal efficiency of the globus pallidus, and BG_R_6_4 corresponds to the weighted nodal efficiency of the ventromedial putamen. Reproduced with permission from Cheng *et al.*, Sci. Rep. **13**, 6958 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution (CC BY) License.[122]

D. Artificial neural networks ML approaches

ANNs are extremely adaptable ML models that mimic the assembly of the human brain to study patterns in sensor-driven data.[130–134] They entail interrelated layers of neurons, comprising an input layer (sensor features), hidden layers (feature transformation), and an output layer (predictions). They consist of interconnected layers of computational nodes ("neurons"). Each neuron computes a weighted sum of its inputs, adds a bias, and then passes the result through a nonlinear activation function, σ. The output, a, of a single neuron is given by

$$a = \sigma(w^T x + b) = \sigma\left(\sum_i w_i x_i + b\right), \quad (14)$$

where x is the input vector, w is the vector of weights, and b is the bias. The activation function [e.g., sigmoid, tanh, or the rectified linear unit (ReLU), $f(z) = \max(0,z)$] is crucial for enabling the network to model nonlinear relationships. The network "learns" by adjusting its weights and biases to minimize a loss function. This is typically done via backpropagation, an algorithm that uses the chain rule of calculus to efficiently compute the gradient of the loss function with respect to every weight, allowing for optimization via methods like stochastic gradient descent. This structure makes ANNs universal function approximators, ideal for applications such as predictive maintenance and environmental forecasting.[130,131,135–140]

ANNs are frequently used to model systems where first-principles equations are incomplete or have unknown parameters.

One such complex system is bacterial infections in hospital environments, where numerous biological, environmental, anthropogenic, and medical factors influence the spread and severity of contagions. Modeling these systems is challenging due to the intricate interactions between microbial populations, host responses, and external conditions, which can be done with the aid of sensing technologies. *Pseudomonas aeruginosa* is a serious infection in hospitals due to its quorum sensing (QS) mechanism, which allows the *P. aeruginosa* to evade recognition by the immune systems of the hosts and only turn damaging when their colonies spread uncontrollably to high populations. While mathematical models using ordinary differential equations (ODEs) aid in explaining QS behavior, the variability in environmental conditions is problematic. Naz and Kashyap[141] integrated ANNs with ODE models, using sensor-driven data to advance QS characteristic prediction. The ANN-based model attained a mean concentration error of 0.0637, precisely assessing 4 out of 11 QS-related coefficients within 40% of their identified values. By improving infection prediction modeling, this AI-driven method supports early diagnosis, treatment planning, and better contagion regulation in healthcare settings.

The nanocomposites are complex systems due to the intricate interactions between the components at molecular and macroscopic levels, governing their physicochemical properties such as electrical conductivity, which are useful in gas sensing applications. In this context, ML models, predominantly ANNs, are adopted to comprehend and predict the behavior of such complex systems. Boubila *et al.*[142] demonstrated the use of the ANN algorithm to model the electrical conductivity and gas sensing performance of PANI/Gr nanocomposites using a vast dataset collected from over 100 references. One of the major results was the identification of critical input parameters that govern the sensing performance of the composite material. The model employed a feed-forward design with multiple hidden layers and was trained by a backpropagation algorithm. The input parameters comprised numerous properties of the nanocomposite material, including the type of additive/precursors, concentration, and temperature, while the output predicted the gas-sensing behavior and conductivity. The model effectively determined that the conductivity profiles of additives had a substantial impact on the gas-sensing efficacy of the composite. Moreover, it addressed the challenge of outliers in the dataset, presenting methods for outlier detection and mitigation, thus confirming the robustness of the ANN models. These results validate the potential of ANN-based models for precisely predicting the performance of complex sensor systems like PANI/Gr nanocomposites and open possibilities for the optimized design of materials for supercapacitors, gas sensors, and energy storage.

The ability of ANNs to handle numerous interacting variables makes them powerful tools for environmental modeling. Urban regions are quickly developing, which necessitates optimizing the use of solar energy, which is critical for attaining sustainability and reducing dependence on fossil fuels. However, predicting solar radiation in metropolises is a complex assignment due to the many dynamic urban factors affecting it. Tehrani *et al.*[143] addressed it by applying an ANN model to predict solar radiation based on numerous urban characteristics, such as geographical coordinates, building heights, land habitation, and azimuth angles. They trained the model using data from 20 different cities and efficiently learned the nonlinear relationships between these factors, attaining a high R^2 value of 85% with marginal prediction error. This sensor-driven data modeling approach allows more precise forecasts of solar radiation, proposing to urban planners the tools required to optimize city infrastructure and endorse sustainable practices.

Modeling the complex interactions that occur between CO_2 uptake and various parameters of activated carbon adsorbent fabrication is important for growing effective materials for CO_2 capture. Mashhadimoslem *et al.*[144] used the ANN model to predict the specific surface area and CO_2 adsorption ability of activated carbon synthesized from diverse biomass types. They collected data from 35 publications, considering multiple input variables, such as precursors, activators, pyrolysis temperature, and adsorption conditions. Four ANN structures were made using multilayer perceptron and radial-based function algorithms, optimized with Bayesian regularization backpropagation. The outcomes exhibited high accuracy ($R^2 > 0.99$), making ANN a favorable tool for forecasting the performance of AC adsorbents in CO_2 capture. This sensor-driven modeling approach helps in designing efficient adsorbents, contributing to sustainable CO_2 capture technologies [Figs. 6(a)–6(d)].

ANNs are powerful tools but possess several limitations, such as high complexity and long training times, particularly for deep models. The requirement for huge computational resources and time can be a substantial challenge, predominantly for real-time or resource-constrained applications. Finally, due to the large number of free parameters (weights), ANNs require massive datasets and significant computational resources to train effectively and avoid overfitting.[145–147]

E. k-nearest neighbor ML approaches

k-nearest neighbor (k-NN) is a non-parametric, instance-based learning algorithm that is utilized for classification and regression. It functions on the principle that similar data readings tend to fit into the same category.[148,149] To make a prediction for a new query point x_q, k-NN identifies the k closest data points (its "neighbors") from the training set, $N_k(x_q)$. The distance is typically measured using a metric like the Euclidean distance between two points p and q in a p-dimensional space,

$$d(p,q) = \sqrt{\sum\nolimits_{i=1}^{p} (p_i - q_i)^2}. \tag{15}$$

During prediction analysis, k-NN determines the k closest data points (neighbors) using a distance metric such as Euclidean distance and allocates the most common category for categorization, or averages the values for regression. Despite its simplicity, k-NN is widely used in anomaly detection, wearable sensor analysis, and environmental monitoring. It requires no explicit training phase, but can be computationally expensive at prediction time.[52,150–154]

In dynamic systems such as smart grids or solar photovoltaic (PV) arrays, k-NN acts as an effective non-parametric forecasting tool. It operates on the physical assumption that similar past system states (in terms of sensor readings) will lead to similar future outcomes. Hasnat and Rahnamay-Naeini[155] employed ML to examine multivariate time series data through instantaneous correlations, recognizing system anomalies due to cyber-attacks and single-line failures. A k-NN classification model, trained on particular features from these correlations, is used for real-time stress monitoring and localization. The outcomes show that this sensor-driven method efficiently

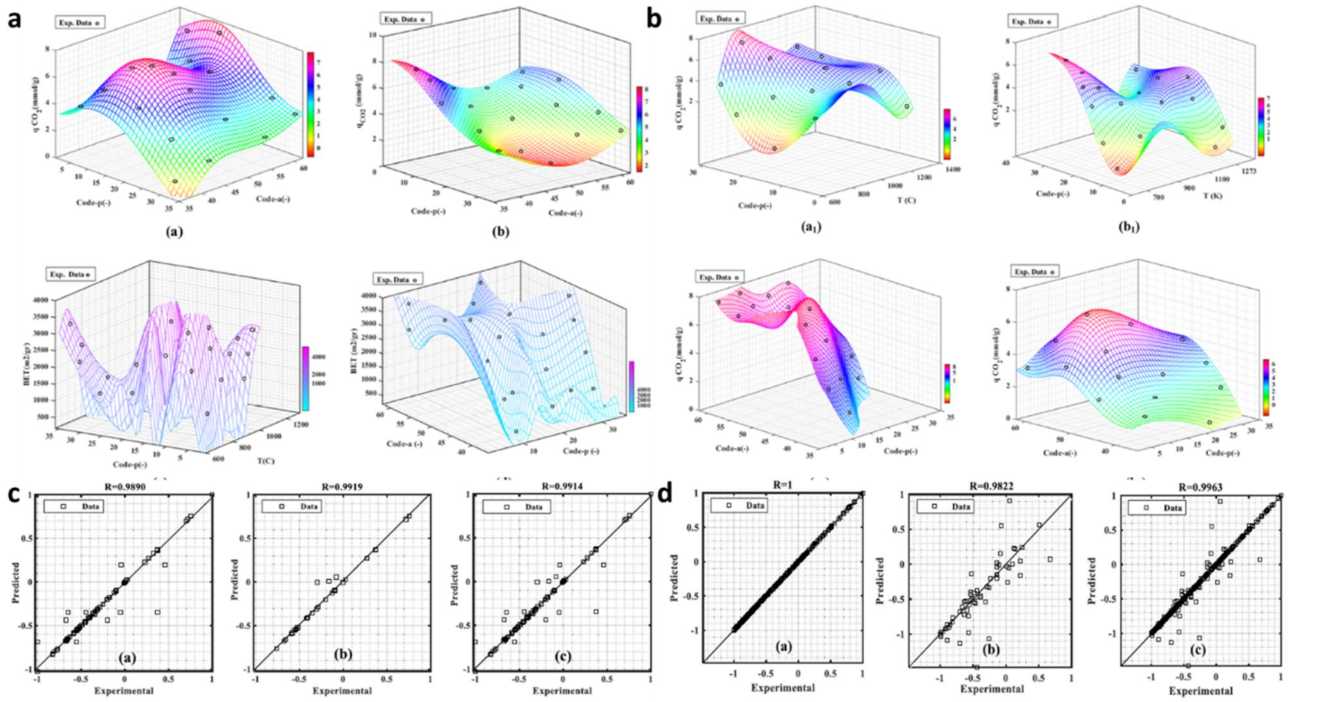

FIG. 6. (a) 3D response surface plots generated using Artificial Neural Networks (ANNs), specifically utilizing Multi-Layer Perceptron (MLP) and radial basis function (RBF) models, aim to predict (a) CO_2 adsorption with MLP, (b) CO_2 adsorption with RBF, (c) BET results using MLP, and (d) BET results using RBF. (b) 3D response surface plots are developed using the ANN-MLP model to forecast CO_2 adsorption-MLP at temperatures of 273 K (a1) and (a2) and 298 K (b1) and (b2). (c) Comparing experimental and predicted BET data using the MLP model for: (a) the training phase, (b) the testing and validation phase, and (c) the complete dataset. (d) Compares experimental and predicted CO_2 adsorption data using the MLP model for: (a) the training phase, (b) the testing and validation phase, and (c) the complete dataset. Reproduced with permission from Mashhadimoslem *et al.*, Ind. Eng. Chem. Res. **60**(38), 13950–13966 (2021). Copyright 2021 American Chemical Society.[144]

improves the security and resilience of smart grids, strengthening the function of ML in complex system monitoring [Figs. 7(a)–7(d)].k-NN can serve as a computationally efficient surrogate model for complex physical simulations. Complex biological sensing methods need very sensitive and selective monitoring mechanisms, particularly for hormone sensing in medical diagnostics. Wekalao *et al.*[156] integrated gold metasurfaces with graphene to construct a terahertz-based sensor that proficiently senses reproductive hormones by observing refractive index variations. The complexity of the system is due to the interaction between nanomaterials, electromagnetic wave interactions, and biological samples. Finite element method simulations were used to optimize design parameters, attaining a sensitivity of 375 GHzRIU^{-1} and a limit of detection of 0.556 RIU. The efficiency was further enhanced by integrating a k-NN regressor model, decreasing computational resources and simulation time by 85% while enhancing predictive accuracy. This sensor-driven ML-assisted method explores the potential of data modeling in optimizing complex biosensing systems for biomedical applications.

Solar photovoltaic systems display intrinsic variability due to changing environmental settings, requiring dynamic and adaptive models for precise performance prediction. A study presented a novel dynamic/adaptive k-NN model to estimate photovoltaic conversion efficacy by recognizing analogous historical functioning conditions, assigning similarity-based weights, and computing a weighted efficacy average.[157] It was applied to a 7.98 kW rooftop solar PV system in Jordan and revealed good predictive accuracy compared with conventional regression models. The projected method improved efficacy by 45.2% and R^2 by 6.8%, making it an effective tool for real-world photovoltaic performance forecasting [Figs. 8(a)–8(f)].

Dopamine sensing is crucial in diagnosing and monitoring neurological disorders like Parkinson's disease and schizophrenia. However, conventional diagnostics possess limitations in sensitivity and real-time analysis due to the complexity of biological cells. Wekalao and Mandela[158] employed an innovative terahertz-based biosensor based on surface plasmon resonance principles for direct dopamine sensing. The sensor was optimized using computational electromagnetic simulations across the 0.1–0.6 THz range, exhibiting high sensitivity (500 GHz/RIU) and a detection limit of 0.867 RIU. Moreover, a k-NN regression model is applied to improve predictive efficacy, attaining an R^2 of up to 1.0. This integration of nanophotonics and ML provides development in rapid, sensitive, and consistent dopamine sensing for clinical applications.

The ability of the algorithm to find patterns in high-dimensional data makes it suitable for analyzing complex human behavioral data. Real-time stress monitoring is critical in healthcare, but conventional physiological monitoring approaches necessitate specialized equipment, making incessant tracking impractical. A study explored a sensor-driven data modeling method that employed smartphone-based

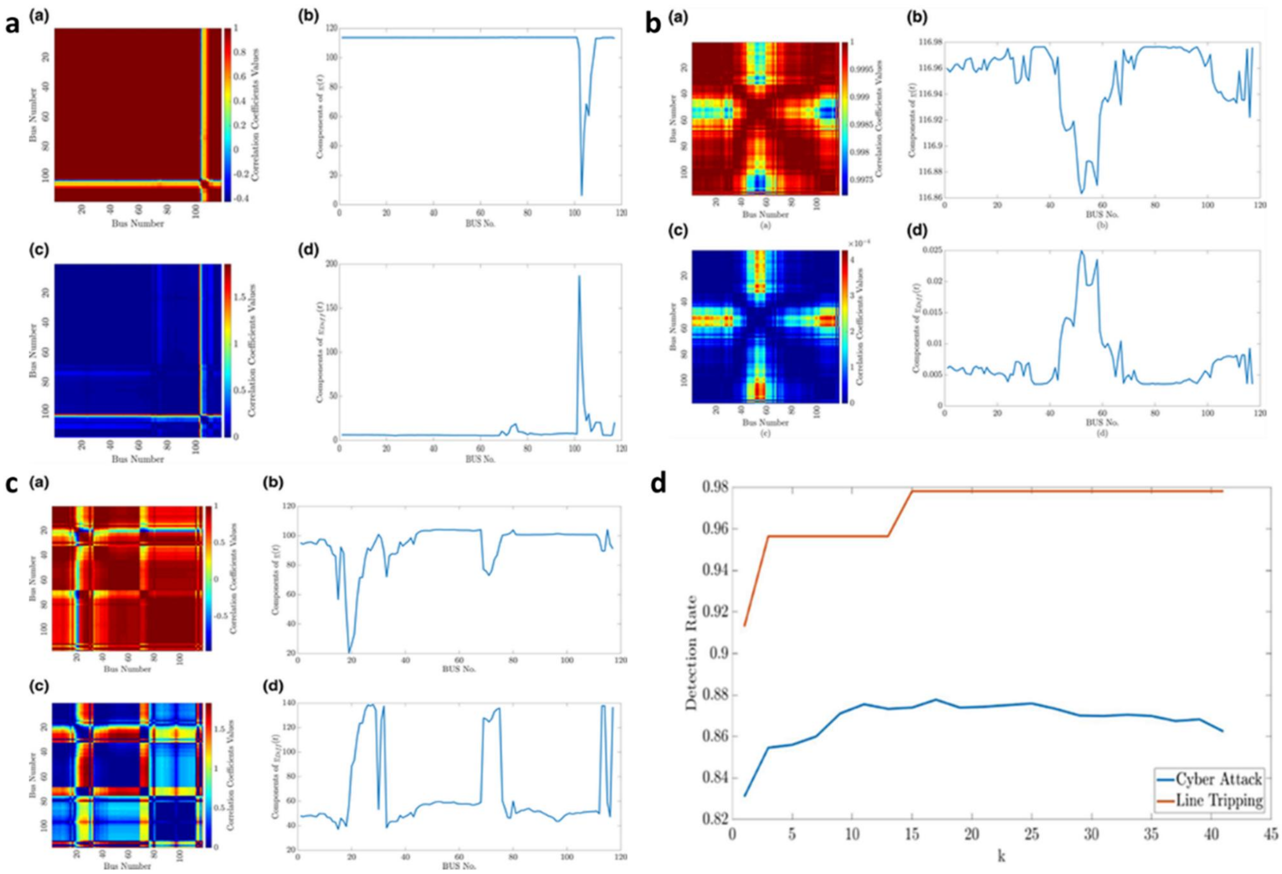

FIG. 7. (a) Investigating how a single line failure impacts (a) the instantaneous correlation matrix image C(t) (b) the aggregated correlation vector, (c) the correlation difference matrix image, CDiff(t), and (d). (b) Considering a scenario with no stress: (a) the instantaneous correlation matrix image C(t), (b) the aggregated correlation vector, (c) the correlation difference matrix image, CDiff(t), and (d). (c) The impact of tripping the transmission line 28, linking bus numbers 21 and 22: (a) the instantaneous correlation matrix image C(t) (b) the aggregated correlation vector, (c) the correlation difference matrix image, Cdiff(t), and (d). (d) Influence of the parameter k on the detection rate in the k-NN method. Reproduced with permission from M. A. Hasnat and M. Rahnamay-Naeini, IET Smart Grid **4**, 307–320 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) License.[155]

behavioral analysis for stress monitoring.[159] By examining soft keyboard typing patterns, data from accelerometers, gyroscopes, gravity sensors, and touchscreen interactions were collected. A total of 172 features were extracted, but feature selection techniques, including filter-based methods and genetic algorithms, were used to increase computational efficacy. Among the tested ML models, k-NN exhibited the utmost accuracy (89.61%) and F-measure (0.9052). Moreover, a mobile application was developed for real-time stress monitoring and personalized relaxation interventions. This study emphasizes the potential of sensor-driven data modeling in healthcare applications, indicating the role of AI-enhanced behavioral analytics to advance stress monitoring and intervention approaches in complex systems.

Despite all its merits, k-NN has various limitations, especially in sensor-driven data modeling for complex systems. It is computationally costly as it needs to store the entire dataset and compute distances for every query, making it ineffective for large datasets. Moreover, it has limitations in dealing with dimensionality, where the performance reduces as the number of features rises due to sparse data distributions. The algorithm is also extremely sensitive to noisy data and irrelevant features, as it relies on raw distance measurements.[160–163]

F. Gradient boosting machines ML approaches

GBMs and their optimized variant, XGBoost, are collective learning procedures that shape manifold weak predictive models (decision trees) and integrate them iteratively to increase accuracy. They are predominantly effective in managing complex sensor data with nonlinear relationships. Unlike random forests, which build trees in parallel, GBMs build them sequentially.[164–167] Each new tree is trained to correct the errors of its predecessor. Formally, a GBM is an additive model where the prediction at step m, Fm(x), is given by

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x). \tag{16}$$

Here, $F_{m-1}(\boldsymbol{x})$ is the model from the previous step, $h_m(\boldsymbol{x})$ is the new weak learner (tree), and γ_m is the learning rate. The key insight of gradient boosting is that the new tree h_m is trained to predict the negative gradient of the loss function *L(y,F(x))* with respect to the previous

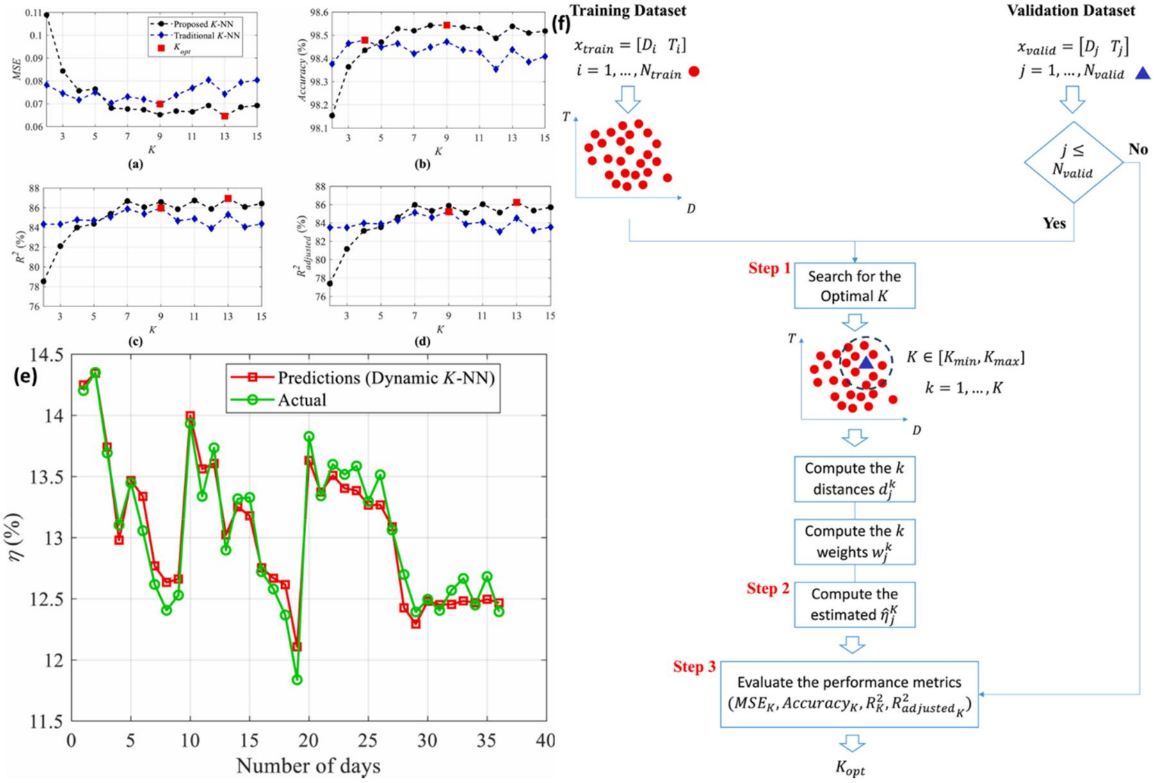

FIG. 8. Evolution of key performance metrics with increasing number of nearest neighbors (NNs) for both conventional and proposed dynamic K-NN prediction models: (a) mean squared error (MSE) (b) prediction accuracy, (c) coefficient of determination (R^2), and (d) adjusted R^2. (e) Comparison between actual and predicted values of photovoltaic (PV) daily conversion efficiency for the test dataset using the proposed dynamic K-NN model. (f) Schematic representation of the proposed data-driven dynamic K-NN framework for PV system conversion efficiency estimation. Reproduced with permission from Al-Dahidi *et al.*, Results Eng. **22**, 102141 (2024). Copyright 2024 Elsevier.[157]

prediction of the model. These negative gradients are often called "pseudo-residuals." GBMs employ boosting methods to diminish errors at every iteration, resulting in significant predictive outcomes.

XGBoost further improves this by integrating regularization and parallel processing, making it rapid and more effective for large-scale sensor-driven data applications. XGBoost improves upon this by minimizing a regularized objective function at each step,

$$\mathrm{Obj}^{(m)} = \sum_{i=1}^{N} \mathcal{L}\left(y_i, \widehat{y_i^{(m-1)}} + f_m(x_i)\right) + \Omega(f_m), \qquad (17)$$

where the regularization term, $\Omega(f_m) = \gamma T + \frac{1}{2}\lambda||w||^2$. These models are widely used in predictive maintenance, failure prediction, and air quality monitoring, where sensor data are frequently noisy and multidimensional.[168–170]

Boosting models are particularly effective at handling the heterogeneous and complex datasets found in environmental science. Toharudin *et al.*[171] studied the application of sensor-driven data modeling in addressing the challenge of air quality determination in Jakarta, one of the world's most polluted cities. Jakarta's $PM_{2.5}$ concentrations peaked at 148 μg/m^3 in 2022, as reported by the Meteorological, Climatological, and Geophysical Agency. However, the collected data showed an unbalanced class distribution, posing a challenge for accurate classification. To address this, a boosting algorithm-based supervised learning method was used to classify $PM_{2.5}$ concentration levels by examining meteorological patterns from 2015 to 2022. The study evaluated adaptive boosting, extreme gradient boosting, categorical boosting, and light gradient boosting machines. The outcomes exhibited that boosting classifiers efficiently mitigated bias and enhanced classification accuracy despite imbalanced data, classifying $PM_{2.5}$ levels into good (62%), moderate (34%), and unhealthy (59%). It highlights the potential of advanced ML techniques in processing sensor-driven environmental data for real-time air quality monitoring and decision-making [Figs. 9(a) and 9(b)].

The high accuracy of boosting makes it suitable for critical diagnostic applications. Mahdizadeh Gharakhanlou and Perez[172] explored sensor-driven data modeling for complex agricultural systems, concentrating on ensemble ML for crop yield prediction in Canada. Accurate yield prediction is crucial for farmers and policymakers, yet the application of advanced ML tools in this domain remains underutilized. This study assesses five ensemble ML models, including gradient boosting machine, AdaBoost, LightGBM, XGBoost, and RF, using different input parameters, including climate data, soil characteristics, satellite-derived vegetation indices, and honeybee census data. Data preprocessing concerned spatial interpolation in ArcGIS Pro and structured formatting in Python. Performance metrics such as root

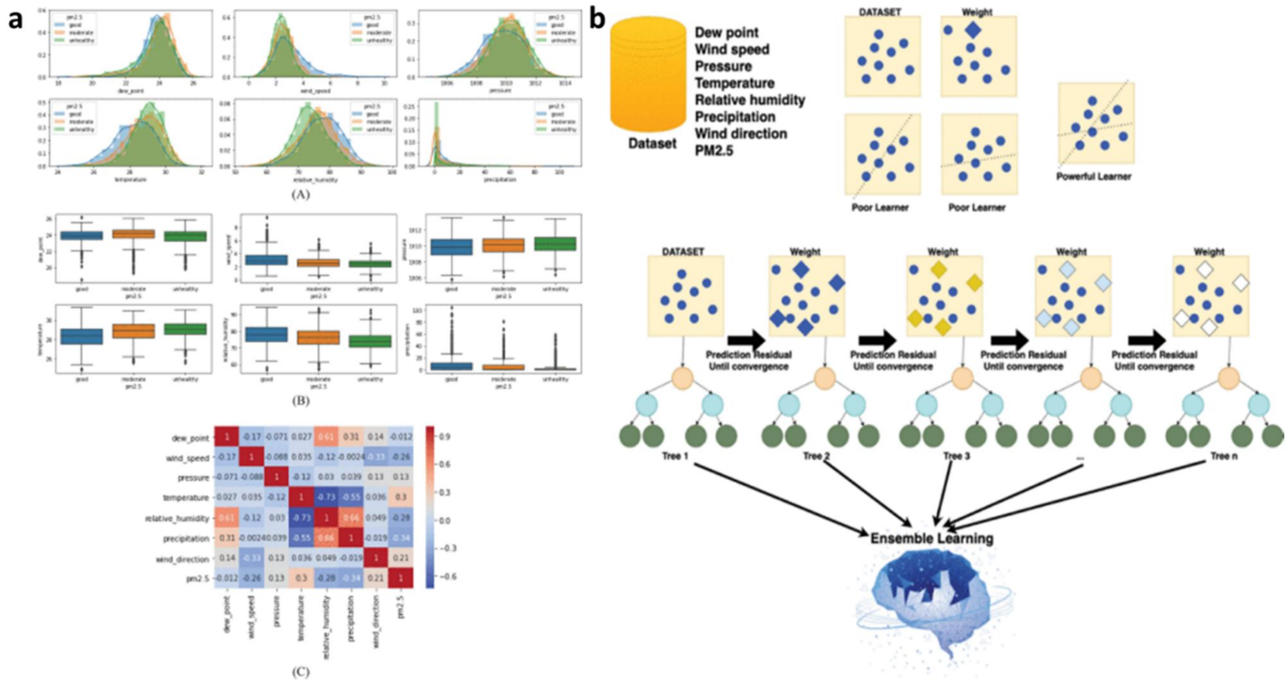

FIG. 9. **(a)** Distribution frequency (A), Boxplot of PM2.5 concentration, and correlation toward PM2.5, (**b**) Data integration toward ensemble learning. Reproduced with permission from Toharudin *et al.* IEEE Access **11**, 35680–35696 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution (CC BY) License.[171]

mean squared error (RMSE), R-squared, and mean absolute error (MAE) identified XGBoost as the most accurate model, achieving an R^2 of 0.95 for canola and 0.96 for soybeans. Honeybee populations were identified as the most influential factor, contributing over 50% to yield variations. The study further examined yield projections under climate change scenarios, revealing significant negative impacts on canola and soybeans. By integrating sensor-driven data with advanced ML modeling, this approach enhances predictive accuracy, supports localized agricultural planning, and provides a framework for climate resilience in farming.

Sensor-driven data modeling in complex systems, which employs an electronic nose for respiratory disease diagnosis, is a popular modern-age diagnostic. Embedded with a chemical gas sensor array, the electric nose captures volatile organic compounds from exhaled breath, converting physiological signals into examinable data. A study involving 199 participants (93 controls, 51 lung cancer patients, and 55 COPD patients) demonstrated the sensor array integration and ML to model the complex biochemical patterns of pulmonary diseases.[173] Among the tested ML models, the XGBoost algorithm attained the utmost classification accuracy of around 79.31% for lung cancer and 76.67% for COPD. The system's compactness, cost-effectiveness, and real-time processing highlight its potential in complex health diagnostics through sensor-driven data modeling in medical applications.

While GBMs give robust predictive performance, their limitations necessitate more advanced and integrated methods. GBMs are computationally expensive, particularly for large datasets, as boosting algorithms form trees successively, resulting in long training periods. They are also prone to overfitting if hyperparameters are not cautiously optimized. Moreover, they struggle with handling missing data proficiently and frequently need extensive feature engineering. These limitations emphasize the requirement for hybrid methods that integrate deep learning, reinforcement learning, and sensor fusion practices to improve scalability, robustness, and real-time adaptability in complex data-driven environments.[174–177]

G. Combined supervised algorithms

Supervised learning procedures provide robust tools for examining and predicting results from sensor-driven data. The choice of method varies with the complexity of the sensor-driven data, computational resources, and the targeted application necessities, ensuring accurate and proficient sensor-based modeling for various complex real-world problems. Supervised learning techniques were also coupled together to overcome each other's demerits and provide improved outcomes.[178–182] While individual supervised learning algorithms are powerful, their capabilities can be enhanced by combining them into hybrid models. These approaches have been considered to leverage the strengths of different algorithms and overcome their individual weaknesses, often leading to superior performance. Two common strategies are stacking and using models as feature extractors. Stacking (or stacked generalization) is an ensemble technique where the predictions from multiple base models (f_1, f_2,…, f_k) are used as input features for a final "meta-model" (g). The final prediction is therefore a function of the outputs of the base models,

$$\widehat{y_{\text{final}}} = g\left(\widehat{y}_1, \widehat{y}_2, \ldots, \widehat{y}_k\right) = g(f_1(x), f_2(x), \ldots, f_k(x)). \tag{18}$$

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Feature engineering pipelines use one model to process raw data and generate high-level features, which are then fed into a second model for the final prediction task. This is common when a complex model is needed for signal processing, but a simpler model is desired for interpretation or classification.

Hybrid models are particularly effective in medical diagnostics where sensor data are complex and accuracy is critical. For instance, Dong *et al.*[183] coupled XGBoost and logistic regression models to design non-contact screening of COVID-19 using sensor-driven data. They used an impulse-radio ultra-wideband radar-enabled sensor to monitor physiological indicators such as respiration, heart rate, sleep quality, and body movement from 23 COVID-19 patients and compared them with healthy subjects. They employed an XGBoost algorithm to process the raw sensor data and developed new and efficient features, which were fed to train an LR algorithm. The coupled supervised ML tool established exceptional discrimination (precision of around 92.5% and a recall rate of around 96.8%), outperforming other solitary ML algorithms. Since contagious diseases have similar inflammatory indicators, the developed model can be used as a facile, cost-effective, automatic screening technique during future infectious outbreaks, especially in underdeveloped countries/regions without adequate healthcare resources. However, the model has several limitations, including a small sample size, slow monitoring capacity of one patient at a time, utilization of continuous-wave radar, and neglect of environmental factors, which need to be addressed to effectively model this complex relationship. Integrating them with physics law and biosensors has also been proven to be effective in healthcare [Figs. 10(a)–10(c)].

Recently, Pathak *et al.*[184] coupled an ANN with logistic regression algorithms on photonic crystal sensors, which are able to detect alterations in the refractive index of biological tissues due to cancer cells for enhanced diagnosis and treatment of thyroid cancer. They reported a two-stage detection procedure in which the first stage is based on a logistic regression algorithm classifying the samples into benign and malignant categories based on the sensor outcomes, and the second stage is based on an ANN algorithm distinguishing thyroid cancer subtypes based on spectral features. They used a dataset containing 120 thyroid samples and attained 91% accuracy at the first stage and 82 % at the second stage, which can be used for noninvasive diagnosis of thyroid cancer.

Combining models can improve the signal-to-noise ratio in advanced sensors and optimize complex agricultural systems. For instance, Tian *et al.*[185] used ML algorithms to improve the spatiotemporal accuracy of optical nanosensor arrays designed using single-walled carbon nanotubes by resolving signal alteration due to environmental noise. Logistic regression, ZeroR, and RF ML algorithms were used to sort and map distinct responses of the sensor pixels, differentiating targeted analyte (dopamine) signals from background noise because of fluid motion and mechanical disturbances. Further, they determined single-pixel features, including entropy, the Laplacian operator, and neighboring pixel correlations for the enhanced identification of reaction pixels even at analyte concentrations less than the limit of detection (LOD). The detection model exhibited a 13-fold surge in LOD, 50% faster detection, with F1 scores larger than 0.9. It allows accurate data-driven modeling of complex sensor systems, making nanosensor-based diagnostics more accurate and consistent in real-world applications.

Effective crop selection is vital for increasing agricultural harvests, yet it is highly reliant on regional agro-climatic complex environments. Rani *et al.*[186] presented an ML-driven method integrating

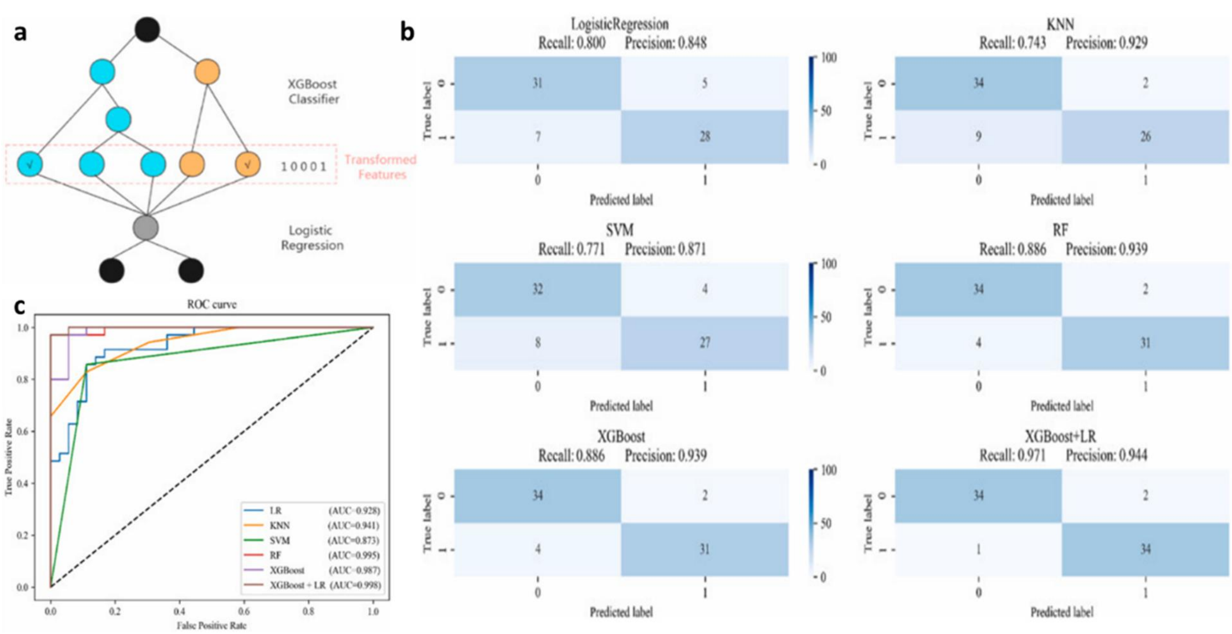

FIG. 10. **(a)** The proposed model integrates XGBoost and logistic regression (LR). Initially, XGBoost serves as a feature selection tool, with its output encoded in one-hot format. Subsequently, these selected features are input into the LR model to produce the final prediction. (b) Confusion matrix for the six models, (c) ROC curves for these models. Reproduced with permission from Dong *et al.*, Comput. Biol. Med. **141**, 105003 (2022). Copyright 2022 Elsevier.[183]

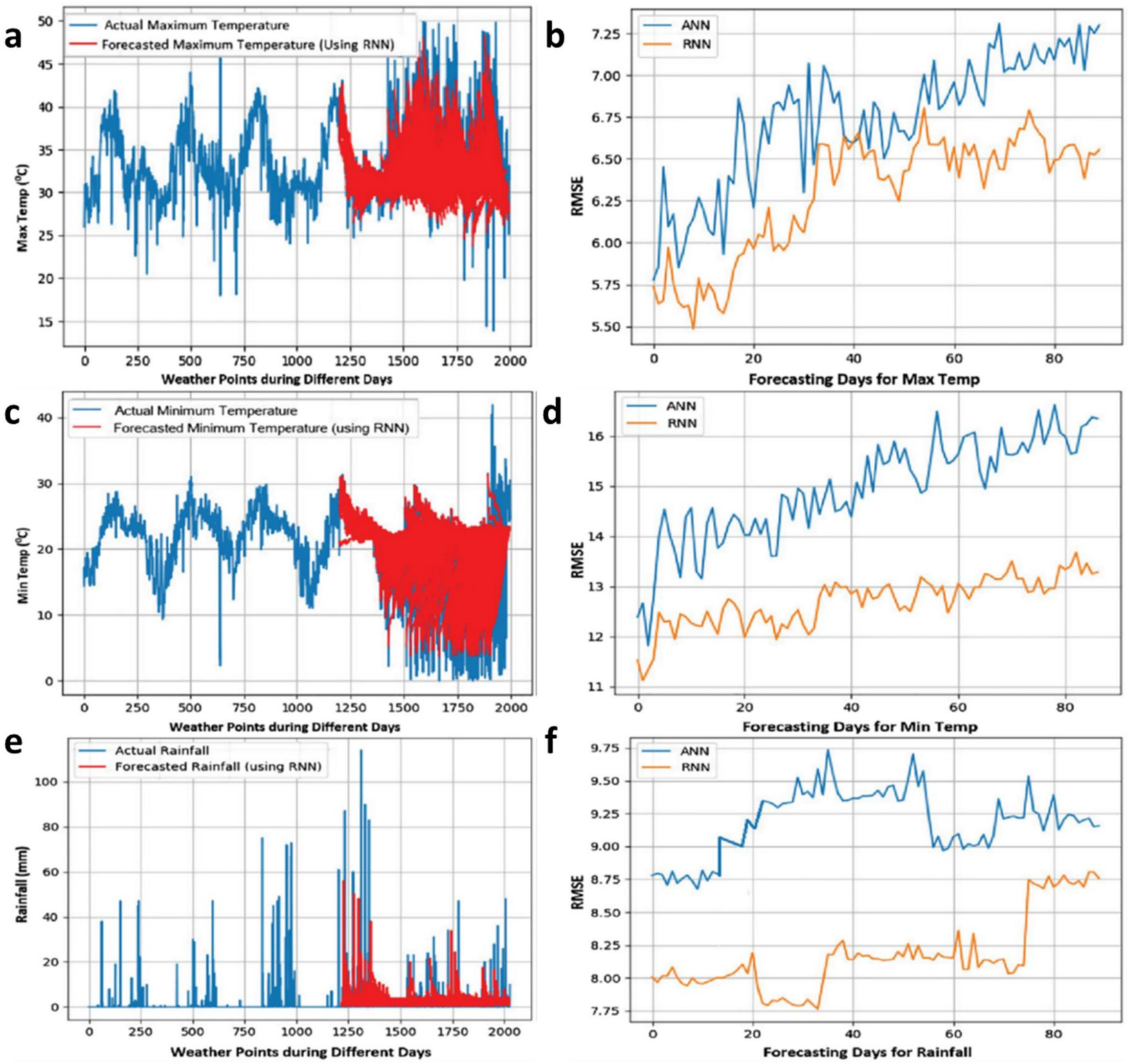

FIG. 11. (**a**) Max temp prediction for 90 days, (b) Max temp day ahead prediction: RMSE value. (c) Min Temp prediction for 90 days. (d): Min temp day ahead prediction: RMSE value. (e) Rainfall prediction for 90 days. (f) Rainfall prediction: RMSE value. Reproduced with permission from Rani *et al.,* Sci. Rep. **13**, 15997 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution (CC BY) License.[186]

LSTM, recurrent neural networks (RNNs) for weather analysis, and an RF classifier for optimal crop selection. The model exhibited excellent performance in weather prediction compared to ANNs, attaining root mean square error values of 5.023% for minimum temperature, 7.28% for maximum temperature, and 8.24% for rainfall prediction. In the second stage, the RF classifier effectively predicted the crop suitability with 97.235% accuracy, resource dependency with 96.437% accuracy, and optimal sowing time with 97.647% accuracy. With a model construction time of only 5.34 s, this methodology provides a fast and data-driven decision-making instrument for farmers, confirming precision agriculture and sustainable resource utilization [Figs. 11(a)–11(f)].

The sensing of volatile organic compounds (VOCs) is intrinsically complex due to cross-sensitivity concerns, necessitating advanced modeling methods for accurate classification and quantification. Nayak *et al.*[187] presented a novel LiFeO micro-flowers VOC sensor with detection capabilities down to 5 ppm for acetone, ethanol, and methanol at 100 °C. The issue of cross-sensitivity was addressed by distinct ML algorithms, including decision tree, RF, kNN, and SVM, which were applied to the response data of the sensor, allowing accurate classification of pure and mixed VOCs. Feature extraction and optimization methods, comprising grid and randomized search, were used to improve the model performance. The best model attained an average classification accuracy of 98% and an R^2-score of 0.98, emphasizing the potential of data-driven approaches in enhancing gas sensor selectivity and reliability for diversified applications.

While efficient in diverse applications, supervised ML algorithms have several shortcomings that necessitate the use of unsupervised approaches for modeling complex systems. Supervised ML depends on large volumes of labeled data, which can be costly and time-consuming to attain in dynamic environments. When used for unknown patterns, these algorithms struggle with simplification, especially in non-stationary/high-dimensional systems where relationships are not definite. Supervised ML is sensitive to class disparities, resulting in biased predictions with skewed data distributions in real-world situations. Moreover, it frequently fails to apprehend hidden structures, hidden variables, or developing patterns that are critical for understanding dynamic complex systems.[188–191]

IV. PHYSICS-INFORMED SUPERVISED LEARNING: FUSING DATA WITH PHYSICAL LAWS

Although data-driven SML models are very efficient at recognizing patterns from empirical data, they do not inherently integrate the foundational physical principles that govern the system of interest. This may lead to predictions that lack physical consistency, require a substantial amount of data for effective generalization, and demonstrate difficulty extrapolating beyond the training domain. These limitations can be addressed using an innovative paradigm of PIML. Within an SML framework, PIML combines known physical laws directly into the training method of the model, usually through adjustments to the loss function. Limited or incomplete labeled data are addressed using data augmentation, semi-supervised and transfer learning strategies, along with the incorporation of unlabeled data. Physics-informed constraints further reduce reliance on labeled samples by guiding the learning process with domain knowledge, enabling physically consistent predictions and improved generalization under sparse data conditions.[28,192–194]

The fundamental idea of physics informed supervised machine learning (PI-SML) is to train a model to fulfill two objectives simultaneously, including to fit the sensor-generated experimental data, and to follow the known physical conservation laws, constitutive relations, or governing boundary conditions. This is done by defining a composite loss function L,

$$\mathcal{L}(\theta) = \mathcal{L}_{data}(\theta) + \lambda \mathcal{L}_{phys}(\theta). \tag{19}$$

Here, $\mathcal{L}_{data}(\theta)$ is the discrepancy between the predictions of the model and the sensor measurements, which is the standard supervised learning loss, such as mean squared error for regression. $\mathcal{L}_{phys}(\theta)$ is a physics-informed regularization term that addresses issues related to the violations of the known physical laws. The hyperparameter λ controls the relative significance of fitting the data while respecting the physical laws. The model's parameters θ, such as the weights and biases of a neural network, are then optimized to reduce this combined loss.

The physics-informed term L_{phys} is expressed as the squared residual of the prevailing partial differential equations (PDEs) or other physical constraints. For a system regulated by a PDE operator P, such as the Navier–Stokes equations for fluid dynamics or reaction-diffusion equations,

$$\mathcal{P}\big(u(x,t), \theta_{phys}\big) = 0, \tag{20}$$

where u denotes the state variables of the system, like velocity, concentration, and θ_{phys} are any physical parameters within the PDEs. The physics-informed loss is then defined by evaluating the PDE residual at a set of collocation points x_c, which may or may not coincide with the sensor,

$$\mathcal{L}_{phys}(\theta) = \frac{1}{N_c} \sum_{j=1}^{N_c} \left|\mathcal{P}\big(u(x_{c,j}, t_{c,j}; \theta), \theta_{phys}\big)\right|^2. \tag{21}$$

This term acts as a robust normalizer, guiding the model to physically consistent solutions, even when sensor data are sparse or noisy. Deep neural networks, generally stated as physics-informed neural networks (PINNs), are predominantly compatible with this framework due to their universal approximation proficiencies and the simplicity of computing gradients of the PDE residuals through automatic differentiation.

In sensor-driven modeling, the PIML framework provides various profound advantages. Its main advantage is a reduced dependence on large, labeled datasets, as the rooted physical laws offer a robust inductive bias, guiding the model to precise solutions even with sparse or noisy sensor data. This physical basis also results in improved generalization and extrapolation, permitting the models to perform consistent predictions for conditions outside the training distribution, which is an analytical failure point for purely data-driven methods. Consequently, PIML confirms that predictions are physically consistent and more interpretable, making the outcomes more constant for scientific and engineering applications. In addition, the framework can be reversed to solve inverse problems, allowing the detection of unknown physical parameters within the governing equations directly from sensor data.[195–200]

Physics-informed SML has been demonstrated to be particularly effective for solving inverse problems in continuum physics. For instance, Raissi *et al.*[201] showed that a neural network constrained by the Navier–Stokes equations can reconstruct entire high-resolution velocity and pressure fields from only a few sensor readings, confirming the solution was physically feasible. Another prevailing application of this PI-SML framework is in environmental geophysics, where PINNs are employed to model subsurface contaminant transport. Ke *et al.*[202] addressed the inverse problem of recognizing unknown parameters in the advection–diffusion equation employing only a small and noisy sensor dataset of contaminant concentration. The PINN model was trained to concurrently match the small sensor data while confirming its predictions that the spatiotemporal concentration field followed the physical constraints. The methodology effectively recognized unknown transport and even nonlinear adsorption parameters with high accuracy, which was very difficult for conventional numerical methods. This validates the effective use of PINNs for system identification and environmental risk assessment, even when challenged with limited and defective data (Fig. 12).

Despite its transformative potential, PI-SML is a developing field with some significant challenges. The main obstacle lies in the optimization process. The composite loss function, which integrates data-fidelity and physics-residual terms, can be challenging to minimize. The gradients from these two terms can have very different magnitudes, resulting in training instabilities where the model may satisfy the data requirements but violate the physics, or vice versa. Their performance is also extremely sensitive to the weighting hyperparameter λ, which needs careful, and often empirical, tuning to accurately balance the impact of the observed sensor data against the physical constraints. In addition, while standard PINNs excel at modeling smooth phenomena in simple domains, they have difficulties related to complex geometries, sharp gradients, or discontinuities like shockwaves in

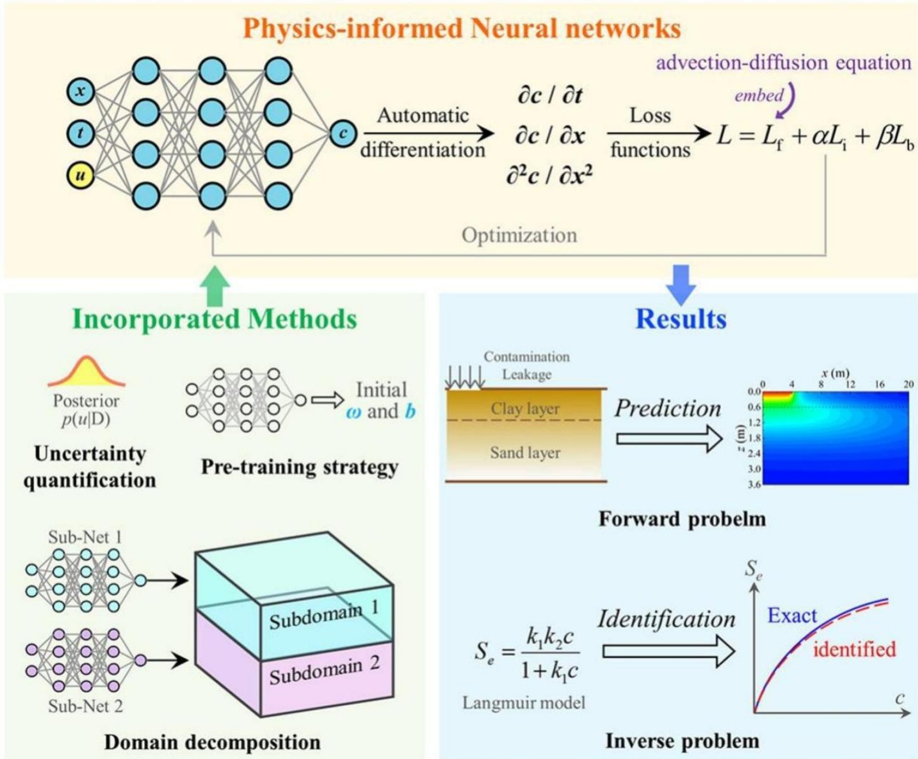

FIG. 12. An illustrative diagram showing a PINN-based deep learning method for examining contaminant transport in soils comprising a pre-trained PINN, a probabilistic PINN model, a domain decomposition strategy, pollutant transport in 2D double-layered soils, a PINN predicted contour map for contaminant concentration, and adsorption isotherm predictions. Reproduced with permission from Ke *et al.*, Comput. Geotech. **180**, 107055 (2025). Copyright 2025 Elsevier.[202]

fluids or fractures in solids, as a single neural network has difficulty indicating such features precisely. They also show a "spectral bias," which makes it challenging to examine high-frequency and multiscale dynamics existing in phenomena such as turbulence. Finally, the practical application of PI-SML frameworks is significantly more complex than conventional supervised learning, necessitating dedicated software libraries and deep expertise in both SML and the underlying physics of the system being modeled.[27,203,204]

V. ADDRESSING CHALLENGES IN SENSOR-DATA-DRIVEN SUPERVISED MODELING OF COMPLEX SYSTEMS AND PROSPECTS

A crucial initial phase to decode any complex system is selecting the appropriate algorithm. Although a universally optimal model does not exist, performing a comparative analysis provides a dedicated direction that is tailored to the distinctive features of the sensor data and the physical context of the problem. For example, tree-based ensembles like RF and XGBoost frequently offer high performance for tabular sensor data with complex interactions. For systems with extremely high dimensionality or where nonlinearities are intense, ANNs are typically better, provided adequate training data are available. Moreover, for systems where a known physical law can be expressed mathematically, PI-SML models provide a better output to achieve high accuracy with a small dataset. The scalability of the framework is ensured through dimensionality reduction, feature selection, and parallel or distributed computing strategies to handle high-dimensional sensor data. Real-time performance is supported via online or incremental learning, while physics-informed constraints reduce the hypothesis space, improving computational efficiency and convergence in large-scale sensor networks.

The integration of concepts from statistics, computer science, and physics is leading to innovative methodologies, which can address the challenges of state-of-the-art SMLs and PI-SMLs:

Quantifying uncertainty with Bayesian methods: Standard SML models stipulate a single-point prediction, providing no sense of confidence, which can be tackled by adopting probabilistic modeling. For instance, Bayesian neural networks and Gaussian processes do not learn a single set of weights but rather a probability distribution over them. It allows them to output not just a prediction but also a reliable interval, which quantifies the uncertainty of the model. This solution is essential for data-heavy fields such as medical diagnostics and structural safety, where model confidence is as crucial as the prediction.

Enhancing trust with interpretable AI (XAI): To solve the "black box" problem of complex models like ANNs and GBMs, the field of eXplainable AI (XAI) is developing. Techniques such as LIME (local interpretable model-agnostic explanations) and SHAP (Shapley Additive exPlanations) are effective methods that can examine a trained model to explain the reason for making a specific prediction based on the sensor inputs. This is a vital solution for creating confidence and enabling scientific discovery.

Overcoming data scarcity with generative and transfer learning: A main panorama for overcoming the "big data" necessity is to use generative models, like generative adversarial networks (GANs) or variational autoencoders (VAEs). These models can learn the fundamental distribution of real sensor data and generate new and physically realistic artificial data points to expand small datasets. Likewise, transfer

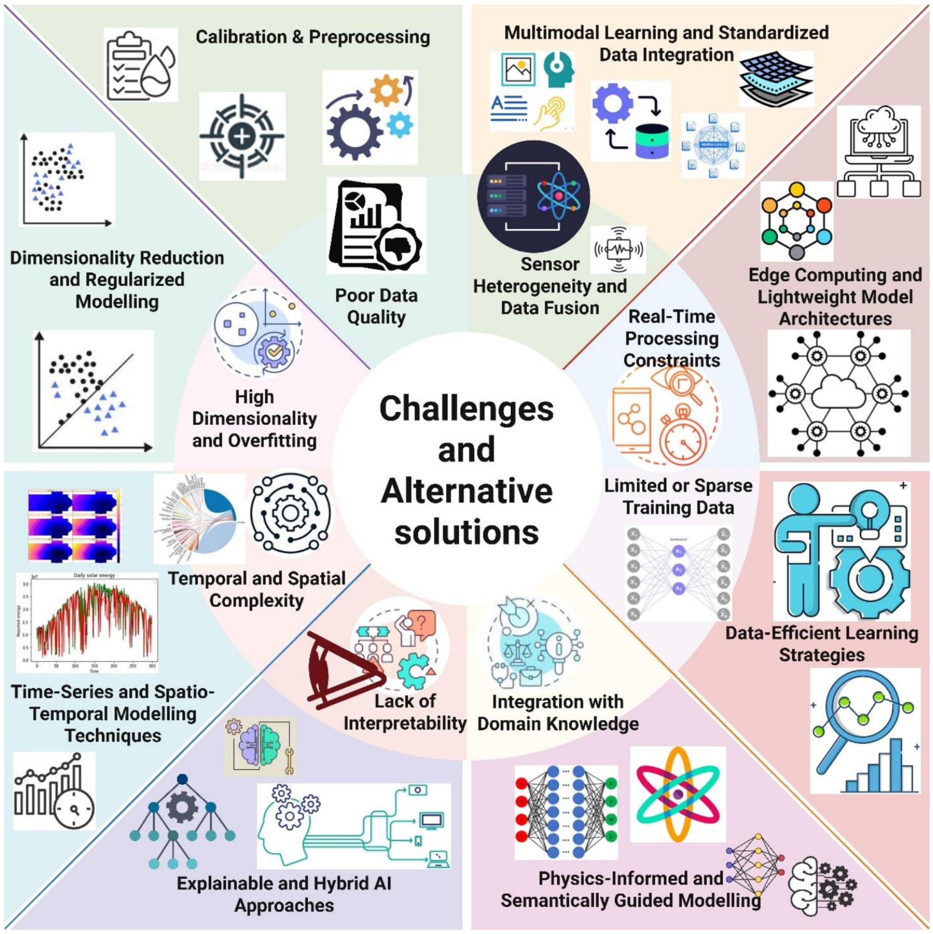

FIG. 13. Overview of major challenges and alternative solutions in sensor-data-driven modeling of complex systems, highlighting strategies for improving data quality, scalability, interpretability, and integration with domain knowledge.

learning, where a model pre-trained on a large and general dataset is modified on a smaller and specific task, is a powerful tool for accelerating learning in low-data environments.

On-sensor intelligence with edge AI: The future of sensing is real-time and on-device processing. Edge AI is an emergent field focused on implementing efficient supervised models directly on sensor hardware. This is attained through solutions such as model quantization (employing lower-precision numbers), pruning (eliminating redundant connections in a neural network), and knowledge distillation (training a small model to mimic a larger one). This prospect will enable a new generation of intelligent and autonomous sensors for applications in robotics, wearables, and the IoT (Fig. 13).

VI. CONCLUSION AND OUTLOOK

The integration of advanced sensor technologies with supervised machine learning is leading a paradigm shift in the ability to model complex physical, biological, and engineered systems. This review has critically examined the primary supervised learning algorithms that convert high-dimensional sensor data into predictive insights. The discussion has considered topics ranging from the basic concepts of classical regression and support vector machines to ensemble methods such as random forests and gradient boosting machines, as well as the nonlinear modeling capabilities of artificial neural networks. It has discussed the importance of data preprocessing and feature engineering to isolate physical signals from noise. Moreover, it highlighted the emergence of PIML as a frontier method, exhibiting how integrating physical laws into the SML algorithms creates models that are not only data-driven but also scientifically grounded, robust, and data-efficient. Although supervised machine learning models perform effectively, they face challenges such as requiring large, labeled datasets, incurring considerable computational costs, and showing complexity that can make their internal functioning difficult to interpret. The path forward lies in adopting emerging methods like Bayesian methods for uncertainty quantification, eXplainable AI for interpretability, and edge AI for on-sensor intelligence. The continued interdisciplinary efforts dedicated to sensor physics, materials science, and supervised learning hold immense potential for developing the predictive, intelligent, and trustworthy solutions required to address pressing global challenges for the betterment of society.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Pradeep Bhadola: Conceptualization (equal); Data curation (equal); Formal Analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing - original draft (equal); Writing - Review & Editing (equal). **Harsh Sable:** Visualization (equal); Writing - original draft (equal). **Pratima Kumari:** Writing - Review & Editing (equal). **Sachin Kadian:** Writing - Review & Editing (equal). **Roger Narayan:** Investigation (equal); Project administration (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing - Review & Editing (equal). **Vishal Chaudhary:** Conceptualization (equal); Data curation (equal); Formal Analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing - original draft (equal); Writing - Review & Editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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